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Toward neuroanatomical and cognitive foundations of macaque social tolerance grades

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This **important** work compares the size of two brain areas, the amygdala and the hippocampus, across 12 species belonging to the *Macaca* genus. The authors find, using a **convincing** methodological approach, that amygdala - but not hippocampal - volume varies with social tolerance grade, with high tolerance species showing larger amygdala than low tolerance species of macaques. Interestingly, their findings also suggest an inverted developmental effect, with intolerant species showing an increase in amygdala volume across the lifespan, compared to tolerant species exhibiting the opposite trend. Overall, this paper offers new insights into the neural basis of social and emotional processing.

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Abstract

The macaque genus includes 25 species with diverse social systems, ranging from low to high social tolerance grades. Such interspecific behavioral variability provides a unique model to tackle the evolutionary foundation of primate social brain. Yet, the neuroanatomical correlates of these social tolerance grades remain unknown. To address this question, we expressed social tolerance grades within a novel cognitive framework and analyzed *post-mortem* structural scans from 12 macaque species. Our results show that amygdala volume is a subcortical predictor of macaques' social tolerance, with high tolerance species exhibiting larger amygdala than low tolerance ones. We further investigated the developmental trajectory of amygdala across social grades and found that intolerant species showed a gradual increase in relative amygdala volume across the lifespan. Unexpectedly, tolerant species exhibited a decrease in relative amygdala volume across the lifespan, contrasting with the age-related increase observed in intolerant species—a developmental pattern previously undescribed in primates. Taken together, these findings provide valuable insights into the cognitive, neuroanatomical and evolutionary basis of primates' social behaviors.

Introduction

Complex social environment implies a greater cognitive demand of social representations and interactions, which is one of the driving forces behind the evolution of the primate brain (Dunbar, 2009; Freeberg et al., 2012; Heldstab et al., 2022). Correlations between social environment and variations in brain structures volumes have been reported, both in humans (Kanai et al., 2012; Maguire et al., 2000; Parkinson et al., 2017) and in non-human primates (NHP) (Noonan et al., 2014a; Sallet et al., 2011; Meguerditchian et al., 2021; Testard et al., 2022).

In rhesus macaques (*M. mulatta*), previous studies have demonstrated that interindividual variation in social characteristics—such as hierarchical status (Noonan et al., 2014b) or group size (Sallet et al., 2011; Testard et al., 2022) – is associated with grey matter volume in core regions of the social brain, including the amygdala, the hippocampus, the superior temporal sulcus (STS), and the rostral prefrontal cortex (rPFC). Supporting the broader relevance of these findings across *Cercopithecinae*, a study in olive baboons (*Papio anubis*) revealed that individuals living in larger social groups exhibited greater total brain volumes, with an effect primarily driven by white matter (Meguerditchian et al., 2021).

Despite the existence of 25 species within the *Macaca* genus (Cooper et al., 2022; Cords, 2012; Ghosh et al., 2022; Thierry, 2007; Thierry et al., 2004), most neuroscience research focuses on two species, *M. mulatta* and *M. fascicularis* (and in rare cases *M. nemestrina* and *M. fuscata* (Carlo et al., 2010; Isa et al., 2009; Maranesi et al., 2014)). In spite of the relatively short evolutionary divergence time within this genus (6 to 8 million years (Perelman et al., 2011)), the various macaque species display a considerable interspecific variety of social behaviors while usually maintaining a multi-male, multi-female, and multi-generational social structure (Balasubramaniam et al., 2017; Thierry, 2007, 2000). These behavioral differences are characterized by different styles of dominance (Balasubramaniam et al., 2012), severity of agonistic interactions (Duboscq et al., 2014), nepotism (Berman and Thierry, 2010; Duboscq et al., 2013; Sueur et al., 2011) and submission signals (De Waal and Luttrell, 1985; Rincon et al., 2023), among the 18 covariant behavioral traits described in Thierry's classification of social tolerance (Thierry, 2021, 2017, 2000).

Despite this large behavioral variability, macaque species display broadly similar general cognitive abilities (ManyPrimates et al., 2022). Specific differences observed in domains such as inhibitory control or social flexibility are thus more likely to reflect adaptive responses to species-specific social constraints, rather than intrinsic disparities in overall intelligence (Joly et al., 2017; Loyant et al., 2023). Altogether, the socio-behavioral diversity within the *Macaca* genus provides a compelling model to investigate how social ecology shapes cognition and its neural substrates.

The concept of social tolerance, central to this comparative approach, has sometimes been used in a vague or unidimensional way. As Bernard Thierry (2021) pointed out, the notion was initially constructed around variations in agonistic relationships – dominance, aggressiveness, appeasement or reconciliation behaviors – before being expanded to include affiliative behaviors, allomaternal care or male–male interactions (Thierry, 2021). These traits do not necessarily align along a single hierarchical axis but rather reflect a multidimensional complexity of social style, in which each trait may have co-evolved with others (Thierry, 2021, 2000; Thierry et al., 2004). Moreover, the lack of a standardized scientific definition has sometimes led to labeling species as “tolerant” or “intolerant” without explicit criteria (Gumert and Ho, 2008; Patzelt et al., 2014). These behavioral differences are characterized by different styles of dominance (Balasubramaniam et al., 2012), severity of agonistic interactions (Duboscq et al., 2014), nepotism (Berman and Thierry, 2010; Duboscq et al., 2013; Sueur et al., 2011) and submission signals (De Waal and Luttrell, 1985; Rincon et al., 2023), among the 18 covariant behavioral traits described in Thierry's classification of social tolerance (Thierry, 2021, 2017, 2000).

To ground the investigation of social tolerance in a comparative neuroanatomical framework, we introduced a tentative working model that articulates behavioral traits, cognitive dimensions, and their potential subcortical neural substrates. Drawing upon 18 behavioral traits identified in Thierry's comparative analyses (Thierry, 2021, 2007), we organized these traits into three core dimensions: socio-cognitive demands, behavioral inhibition, and the predictability of the social environment (Table 1). This conceptualization did not aim to redefine social tolerance itself, but rather to provide a structured basis for testing neuroanatomical hypotheses related to the volume of relevant subcortical areas and social style variability. It echoes recent efforts to bridge

behavioral ecology and cognitive neuroscience by linking specific mental abilities – such as executive functions or metacognition – with distinct prefrontal regions shaped by social and ecological pressures (Bouret et al., 2024 [↗](#); Testard et al., 2022 [↗](#)).

Navigating social life in primate societies requires substantial cognitive resources: individuals must not only track multiple relationships, but also regulate their own behavior, anticipate others' reactions, and adapt flexibly to changing social contexts. Taken advantage of databases of magnetic resonance imaging (MRI) structural scans, we conducted the first comparative study integrating neuroanatomical data and social behavioral data from closely related primate species of the same genus to address the following questions: To what extent can differences in volumes of subcortical brain structures be correlated with varying degrees of social tolerance? Additionally, we explored whether these dispositions reflect primarily innate features, shaped by evolutionary processes, or acquired through socialization within more or less tolerant social environments.

The first category, socio-cognitive demands, refers to the cognitive resources needed to process, monitor, and flexibly adapt to complex social environments. Linking those parameters to neurological data is at the core of the social brain theory (Dunbar, 2009 [↗](#)). Macaques' social systems require advanced abilities in social memory, perspective-taking, and partner evaluation (Freeberg et al., 2012 [↗](#)). This is particularly true in tolerant species, where the increased frequency and diversity of interactions may amplify the demands on cognitive tracking and flexibility. Tolerant macaque species typically live in larger groups with high interaction frequencies, low nepotism, and a wider range of affiliative and cooperative behaviors, including reconciliation, coalition-building, and signal flexibility (Thierry, 2021 [↗](#), 2000 [↗](#)). Tolerant macaque species also exhibit a more diverse and flexible vocal and facial repertoire than intolerant ones which may help reduce ambiguity and facilitate coordination in dense social networks (Rincon et al., 2023 [↗](#); Scopa and Palagi, 2016 [↗](#); Rebout 2020). Experimental studies further show that macaques can use facial expressions to anticipate the likely outcomes of social interactions, suggesting a predictive function of facial signals in managing uncertainty (Micheletta et al., 2012 [↗](#); Waller et al., 2016 [↗](#)). Even within less tolerant species, like *M. mulatta*, individual variation in facial expressivity has been linked to increased centrality in social networks and greater group cohesion, pointing to the adaptive value of expressive signaling across social styles (Whitehouse et al., 2024 [↗](#)).

The second category, inhibitory control, includes traits that involve regulating impulsivity, aggression, or inappropriate responses during social interactions. Tolerant macaques have been shown to perform better in tasks requiring behavioral inhibition and also express lower aggression and emotional reactivity than intolerant macaques both in experimental and in natural contexts (Joly et al., 2017 [↗](#); Loyant et al., 2023 [↗](#)). These features point to stronger self-regulation capacities in species with egalitarian or less rigid hierarchies. More broadly, inhibition – especially in its strategic form (self-control) – has been proposed to play a key role in the cohesion of stable social groups. Comparative analyses across mammals suggest that this capacity has evolved primarily in anthropoid primates, where social bonds require individuals to suppress immediate impulses in favour of longer-term group stability (Dunbar and Shultz, 2025 [↗](#)). This view echoes the conjecture of Passingham and Wise (2012) [↗](#), who proposed that the expansion of lateral prefrontal area BA10 in anthropoids enabled the kind of behavioural flexibility needed to navigate complex social environments (Passingham et al., 2012 [↗](#)).

The third category, social environment predictability, reflects how structured and foreseeable social interactions are within a given society. In tolerant species, social interactions are more fluid and less kin-biased, leading to greater contextual variation and role flexibility, which likely imply a sustained level of social awareness. In fact, as suggested by recent research, such social uncertainty and prolonged incentives are reflected by stress-related physiology: tolerant macaques such as *M. tonkeana* display higher basal cortisol levels, which may be indicative of a chronic mobilization of attentional and regulatory resources to navigate less predictable social environments (Sadoughi et al., 2021 [↗](#)).

Social trait	Underlying social consequences	Cognitive dimension	Neural correlate
Rate of reconciliation	Demands in recognizing partners, recalling social history and regulate affiliation ¹ .	Higher socio-cognitive demands	Amygdala volume higher ^{7,8,9} ; Hippocampus volume higher ^{9,10,11}
Level of complexity of the communication system	Demands in interpreting social signals and adjusting communication to context ² .		
Development of male-to-male coalitions	Demands in social knowledge and strategic social decision ^{3,4} .		
Development of cooperative behaviors	Demands in understanding intentions and coordinating actions during interactions ^{5,6} .		
Intensity of aggression	Demands in inhibiting impulsive behaviors and regulating emotional responses ^{12,13} .	Better inhibitory control	Amygdala volume lower ^{8,17} ; Hippocampus volume unchanged ¹⁷ .
Degree of confidence of social play	Demands in adjusting behavior and inhibiting responses in mutual interactions ^{14,15} .		
Resource distribution evenness	Demands in regulating emotions and adjusting behavior during competitive interactions ¹⁶ .		
Kin bias (nepotism)	Kin knowledge is less informative to predict social relationships ^{18,19} .	Lower predictability of social environment	Amygdala volume higher ^{7,8,9,17} ; Hippocampus volume lower ^{29,30,31} .
Degree of dominance asymmetry	Conflicts are not always won by dominant individuals, making social outcomes less predictable ^{16,20} .		
Development of formal submission signals	Social communication during conflict is less informative to predict conflict outcomes ^{21,22} .		
Intensity of female rank inheritance	Matrilinear knowledge is less informative to predict social relationship ^{23,24} .		
Rate of affiliative contact	Affiliative network is denser and therefore harder to predict ^{25,26,27} .		
Rate of counter-aggression	Subordinates are more likely to retaliate making social outcomes less predictable ^{3,28} .		
Rate of immature interference in mating	Mounting behaviour trigger more social interactions which induce 'erratic' interaction ¹⁴ .	Unclassified	/
Degree of mother protectiveness	Recognizing individuals and anticipating social risks in interaction ³² .		
Development of allomothering behavior	Understanding social roles and interpreting socio-affective cues ³³ .		
Delayed male dispersal	Navigating hierarchies through flexible behavior and inhibition ¹⁶ .		
Centrality of top-ranking males	Tracking relationships and managing distributed social networks ³⁴ .		

1. Thierry and Sapolsky, 2000; 2. Liebal et al., 2014; 3. Petit et al., 1997; 4. Silk, 1999; 5. Demaria and Thierry, 2001; 6. Micheletta et al., 2012; 7. Bickart et al., 2011; 8. Sallet et al., 2011; 9. Testard et al., 2022; 10. Kanai et al., 2012; 11. Todorov et al., 2019; 12. Adams et al., 2015; 13. Loyant et al., 2023; 14. Petit et al., 2008; 15. Scopa and Palagi, 2016; 16. Thierry, 2007; 17. Tottenham et al., 2010; 18. Silk, 2002; 19. Silk et al., 2003; 20. de Vries et al., 2006; 21. Flack et al., 2006; 22. Waller et al., 2013; 23. Hill and Okayasu, 1995; 24. Kutsukake, 2000; 25. Duboscq et al., 2013; 26. Massen et al., 2010; 27. Silk et al., 2003; 28. Balasubramaniam et al., 2012; 29. Kim et al., 2015; 30. Lyons et al., 2001; 31. Meyer and Hamel, 2014; 32. Maestripietri, 1994; 33. Fairbanks, 1990; 34. Sueur et al., 2011;

Table 1. Cognitive and neuroanatomical categorization of behavioral traits associated with macaque social tolerance

Each behavioral trait was individually evaluated based on existing empirical literature regarding the types of cognitive operations it likely involves. When a primary cognitive dimension could be identified, the trait was assigned accordingly. However, some behaviors – such as maternal protection, allomaternal care, or delayed male dispersal – do not map neatly onto a single cognitive process. These traits likely emerge from complex configurations of affective and social-motivational systems, and may be better understood through frameworks such as attachment theory (Suomi, 2008 [↗](#)), which emphasizes the integration of social bonding, emotional regulation, and contextual plasticity. While these dimensions fall beyond the scope of the present framework, they offer promising directions for future research, particularly in relation to the hypothalamic and limbic substrates of social and reproductive behavior.

Rather than forcing these traits into potentially misleading categories, we chose to leave them unclassified within our current cognitive framework. This decision reflects both a commitment to conceptual clarity and the recognition that some behaviors emerge from a convergence of cognitive demands that cannot be neatly isolated. This tripartite framework, leaving aside reproductive-related traits, provides a structured lens through which to link behavioral diversity to specific cognitive processes and generate neuroanatomical predictions.

We therefore associated these three categories with neuroanatomical hypotheses regarding variations in the volume of two subcortical structures of interest, the amygdala and the hippocampus. Based on existing literature on the effects of the socio-cognitive demands, inhibitory control and social unpredictability – particularly when it induces sustained activation of stress-related systems – on the volume of these two regions (Bickart et al., 2011 [↗](#); Caetano et al., 2021 [↗](#); Coplan et al., 2014 [↗](#); Haley et al., 2012 [↗](#); Howell et al., 2014 [↗](#); Lyons et al., 2001 [↗](#); Sallet et al., 2011 [↗](#)), we hypothesized that increased cognitive demands from the social environment could lead to a differential effect on amygdala and hippocampus volumes (Bickart et al., 2011 [↗](#); Haley et al., 2012 [↗](#); Lyons et al., 2001 [↗](#); Sallet et al., 2011 [↗](#)). We summarized our working hypotheses into a table comparing grade 4 and grade 1 species (Table 1 [↗](#)).

This table summarizes the 18 behavioral traits used to characterize social tolerance grades in the *Macaca* genus, based on Thierry's comparative framework. Each trait is associated with a description of its underlying cognitive implications and assigned—when applicable—to one of three cognitive dimensions: (i) socio-cognitive demands (e.g., tracking partners, coordinating actions), (ii) behavioral inhibition (e.g., regulating impulsivity), or (iii) predictability of the social environment (e.g., anticipating interaction outcomes). The final column presents the hypothesized effects of these dimensions on the volume of two subcortical structures: the amygdala and the hippocampus. Traits that could not be clearly assigned to a specific cognitive dimension—often related to maternal or reproductive strategies—are marked as “unclassified”. This framework is used to generate testable predictions about the neural substrates of social style diversity in macaques.

We tested our hypothesis using 43 *post-mortem* MRI acquisitions of 12 macaque species representing the four grades of social tolerance. The dataset was both composed of samples from open access databases (Milham et al., 2018 [↗](#); Navarrete et al., 2018 [↗](#); Sakai et al., 2018 [↗](#)) as well as newly and unpublished samples from the collection of the Centre de Primatologie de l'Université de Strasbourg (CdP) and INSERM-Oxford University. These samples include brain images of *M. tonkeana* and *M. thibetana*, two macaque species that have never been scanned before as well as scan of *M. nigra* that is rare in the existing literature (Navarrete et al., 2018 [↗](#); Sakai et al., 2023 [↗](#)). Up to this date, only one study has included tolerant species of macaque monkeys in such neuroanatomical comparative framework (Jones et al., 2021 [↗](#)). While Jones et al. (2020) identified interspecific differences in amygdala microstructure and serotonergic innervation, their histological approach did not assess structural volumes at the whole-brain level. To our knowledge, our study is the first to report neuroanatomical correlates of social tolerance grades of the *Macaca* genus based on *post-mortem* MRI volumetric analysis. This approach reveals two key findings. First, across species, amygdala volume is positively correlated with social tolerance grades, with more tolerant macaque species exhibiting larger amygdala volumes. Second, developmental trajectories of the amygdala diverge according to social style: in intolerant

species, amygdala volume increases with age – as commonly reported in the literature (Schumann et al., 2019) – whereas in tolerant species, we observe an unexpected marked decrease over the lifespan. This study offers a novel and valuable perspective by comparing inter-species brain structures to investigate the functioning of the social brain, while accounting for key socio-cognitive variables.

Results

We obtained structural MRI scans of 43 macaques of 12 macaque species. Using a semi-automated registration to an atlas (SARM, Hartig et al., 2021), we extracted amygdala and hippocampus volumes and analyzed whether these covaried with social grade and age, using a Bayesian model. The raw relations between the main response variables (the amygdala's and hippocampus volumes) are depicted in Figure 1.

Model quality and coefficients

The R^2 coefficient of determination of the model indicated a large proportion of variability accounted for by the model (90% credible interval: [0.87, 0.97]). The effect of sex was minimal for the amygdala but more pronounced for the hippocampus (Figure 1, 2), whereas husbandry had a limited effect on both regions of interest. Amygdala volume increased with social grade (independently of its interaction with age) and with age (independently of its interaction with social grade). However, the interaction between social grade and age suggested that the trajectory of amygdala volume over the lifespan differs across social grades, as detailed below. Total brain volume was included as a covariate in the model to account for interindividual differences in brain size. For descriptive purposes, its distribution across social grades is shown in Figure S1.

Despite limited sample size, the interaction between social grade and age suggested a differential trajectory of amygdala volume across the lifespan among different social grades (Figure 1, 2).

To further assess group differences, we implemented Bayesian hypothesis testing using a Region of Practical Equivalence (ROPE, (Kruschke, 2015)) approach, with the ROPE defined as $\pm 0.1 \times \sigma$. This method allows classification of results into three categories: (a) a credible difference if the entire posterior interval lies outside the ROPE; (b) an absence of difference if it lies entirely within the ROPE; and (c) inconclusive if it overlaps the ROPE. For the amygdala, social grade 4 (SG4, i.e. tolerant) individuals had credibly larger volumes than social grade 1 (SG1, i.e. intolerant) individuals up to 19 years of age. For the hippocampus, the posterior distribution of the SG4–SG1 difference briefly exceeded the ROPE between approximately 13 and 18 years of age, indicating a credible difference in this age window. Outside this range, the intervals overlapped the ROPE, resulting in inconclusive evidence. However, the 90% posterior intervals remained entirely above zero at all ages, indicating that SG1 individuals never had larger hippocampal volumes than SG4 (Figure 3).

Predicted data

Figure 4 illustrates how amygdala's volume development varies with an individual's social grade over their lifespan. Two results stand out: first, individuals in Social Grade 1 showed a distinct pattern of amygdala volume development compared to other social grades. Although Grade 1 individuals had a smaller amygdala volume in early years compared to the other Social Grades, the amygdala's volume variation slope was steeper than for the other Grades (slope with 90% credible intervals [0.6, 11.0]). This increase contrasted with trends observed in Grades 3 (slope with 90% CI [−7.6, −0.9]) and 4 (slope with 90% CI [−8.0, 1.9]), which showed a decrease in volume over time. Individuals in Grade 2 also showed a slight increase in amygdala volume (slope with 90% CI [−4.4, 9.3]), similar to grade 1 but not as steep.

When comparing Grade 1 and Grade 4, individuals in Grade 4 showed larger amygdala volumes until approximately 19 years of age (Figure 4).

Figure 1. Model predictors of the amygdala and hippocampus, and volume predictions across social tolerance grades.

First row (A–D): Model predictors and responses for amygdala volume. The volume ratio is calculated as the amygdala volume divided by the total brain volume (excluding the myelencephalon and cerebellum). (A) Distribution of amygdala volume ratios across social tolerance grades. (B) Distribution of amygdala volume ratios by sex. (C) Distribution of amygdala volume ratios by husbandry condition (enclosed vs. semi-free). (D) Distribution of amygdala volume ratios by the frozen status. (E) Distribution of amygdala volume ratios by age. Second row (F–J): Model predictors and responses for hippocampal volume. The volume ratio is calculated as the hippocampal volume divided by the total brain volume (excluding the myelencephalon and cerebellum). (F) Distribution of hippocampal volume ratios across social tolerance grades. (G) Distribution of hippocampal volume ratios by sex. (H) Distribution of hippocampal volume ratios by husbandry condition (enclosed vs. semi-free). (I) Distribution of hippocampal volume ratios by the frozen status. (J) Distribution of hippocampal volume ratios by age. Panels A–E et F–J share the same y-axis.

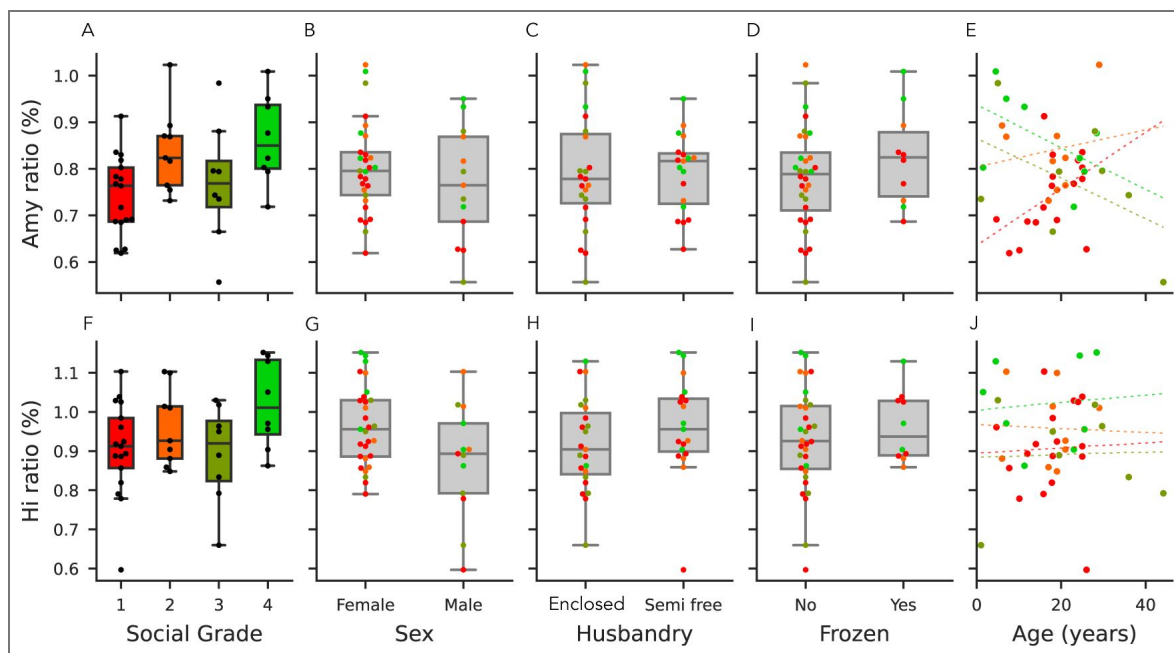
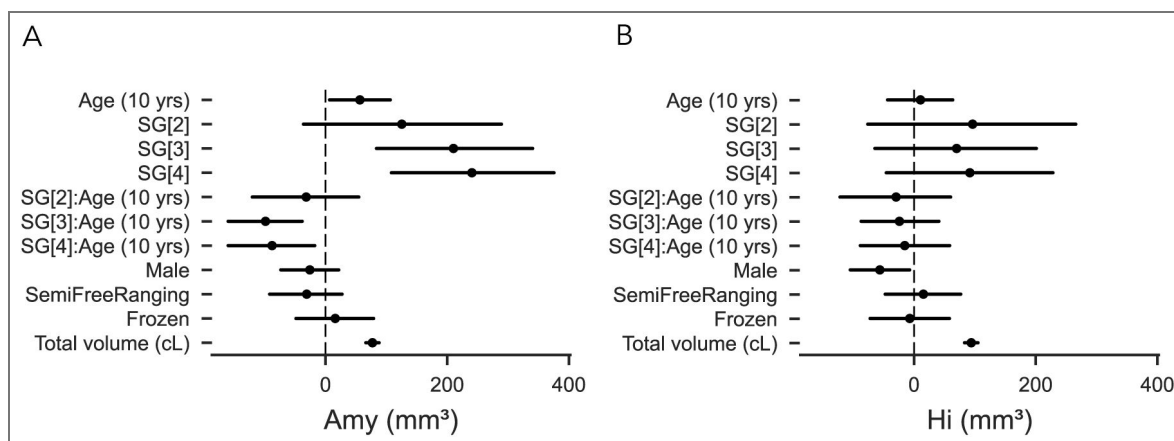


Figure 2. Parameters of the model.

(A) Parameters of the model for the amygdala volume. (B) Parameters of the model for the hippocampal volume. SG [x]: Social Grade [x] vs Social Grade [1]; SG[x]:Age (10 years): Social Grade-Age interaction.



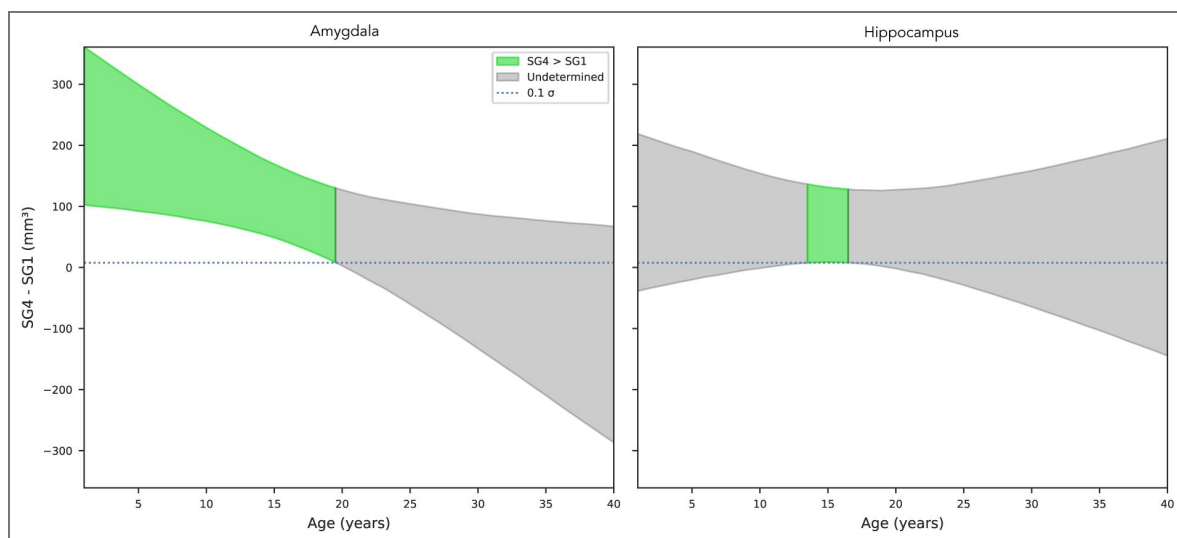


Figure 3. Bayesian hypothesis testing using a Region of Practical Equivalence (ROPE) to assess volume (in mm^3) differences between Social Grade 4 (SG4; tolerant) and Social Grade 1 (SG1; intolerant) across age, for the amygdala (left) and hippocampus (right).

Curves represent median posterior estimates, and shaded areas show 90% credible intervals. Gray bands indicate the ROPE ($\pm 0.1\sigma$). For the amygdala, the difference is credible until ~19 years. For the hippocampus, a credible effect is observed only between ~13 and 18 years.

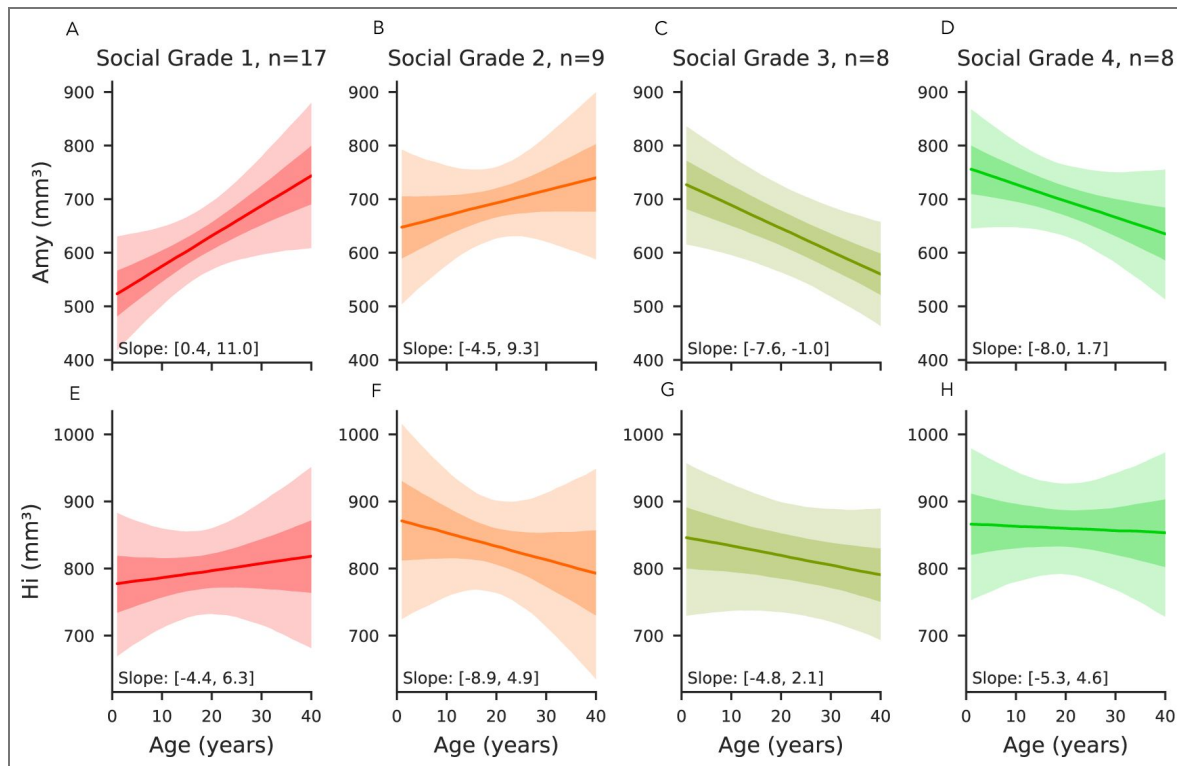


Figure 4. Volume predictions across social tolerance grades of the amygdala and hippocampus.

All panels represent the predictions of the multivariate Bayesian linear model, where all the variables are kept constant (including total brain volumes) in order to represent the effect of age only on the volume of amygdala and hippocampus in mm^3 . First row (A–D): Predicted amygdala volume across social tolerance grades over the lifespan. (A) Predicted amygdala volume as a function of age for grade 1 (intolerant) individuals. (B) Predicted amygdala volume as a function of age for grade 2 individuals. (C) Predicted amygdala volume as a function of age for grade 3 individuals. (D) Predicted amygdala volume as a function of age for grade 4 (tolerant) individuals. Second row (E–H): Predicted hippocampal volume across social tolerance grades over the lifespan. (E) Predicted hippocampal volume as a function of age for grade 1 individuals. (F) Predicted hippocampal volume as a function of age for grade 2 individuals. (G) Predicted hippocampal volume as a function of age for grade 3 individuals. (H) Predicted hippocampal volume as a function of age for grade 4 individuals. In the plots, the solid lines represent the mean predicted values, and the shaded areas indicate the 90% credible intervals, with each social grade shown in a distinct color.

As expected from model predictions, hippocampal volume showed limited variation across age and social grades. ROPE-based hypothesis testing revealed that hippocampal volume in SG1 individuals was never greater than in SG4 individuals, supporting a consistent asymmetry in favor of more tolerant species.

Discussion

We studied for the first time the neuroanatomical foundation of the naturally observed diversity of behavioral traits within the *Macaca* genus. We have assembled a unique database representing nearly half of the known macaque species, with a variety of ages, sexes, and origins. 12 species of them had never been scanned prior to our study. Our investigation focused on the subcortical structures of the brain and more especially the amygdala. This set of nuclei sometimes referred to as a hub of brain networks related to sociality and their social lives are well known for their roles in the stress response (Hölzel et al., 2010 [↗](#)), emotional regulation and social cognition (Noonan et al., 2014b [↗](#); Bickart et al., 2014 [↗](#); Amaral et al., 2003 [↗](#)). Based on *post-mortem* MRI acquisitions from 12 of the 25 macaque species, we showed that amygdala volume correlated with the social tolerance grade and increased with the level of the social grade. Secondly, grade 4 species had a significantly higher amygdala volume at the start of their lives, which decreased over time, compared with grade 1 species, which showed the opposite trend. Finally, further hypothesis testing suggested that species of grade 1 never exhibited larger hippocampus and may have smaller hippocampus around age 15 when compared to those of grade 4. In accordance with our hypotheses (Table 1 [↗](#)), our findings substantiated the assertions that (i) social tolerance is rooted in neuroanatomical differences that can be detected at an early stage of individuals' development, (ii) social styles exert differential influence on subcortical structures throughout individuals' lifespan and (iii) such phenomenon should be mainly driven by the socio-cognitive demands that vary with species social style (as evidenced by the higher amygdala and hippocampus volumes in higher tolerance species).

A neuroanatomical account for social tolerance differences

The social tolerance grades are based on previous ethological observations of behaviors across different species of the genus (Thierry, 2021 [↗](#), 2007 [↗](#)). From these observations, we identified three major cognitive processes—socio-cognitive demands, social predictability, and inhibitory control (see Table 1 [↗](#))—that underpin the observed behaviors. Among the behaviors we classified with “high social-cognitive demand”, several have been previously described in the literature as particularly discriminating between grades 1 and 4. It included greater social network density in grade 4 species (a consequence of, *inter alia*, low nepotism in tolerant species, facilitating interactions between unaffiliated conspecifics) (Balasubramaniam et al., 2017 [↗](#); Sueur et al., 2011 [↗](#)), more complex facial mimics as well as a more complex communication system (Dobson, 2012 [↗](#); Rincon et al., 2023 [↗](#); Scopa and Palagi, 2016 [↗](#); Zannella et al., 2017 [↗](#)), a significantly higher rate of reconciliation (Thierry, 2007 [↗](#)), and a higher frequency of cooperative behaviors, including male-to-male coalition behaviors (De Waal and Luttrell, 1989 [↗](#); Thierry et al., 2008 [↗](#)) in grade 4 species.

At first glance, ones may presume that species of lower social tolerance level, that displayed more overall aggressive behaviours, would have a larger amygdala when compared to more tolerant species. In fact, amygdala ablation or activity modulation in macaque monkeys showed that animals displayed less aggressive and different patterns of social behaviours (Amaral, 2002; Bliss-Moreau 2013, Wellman et al., 2016, Forcelli et al., 2016, Raper et al, 2019), supporting the idea that amygdala activity can promote aggressive behaviours. Our study revealed an opposite trends, amygdala was found to be larger in more tolerant species, and this apparent contradiction invites a more integrative view of the amygdala (Adolphs, 2009 [↗](#); Amaral et al., 2003 [↗](#); Pessoa, 2010 [↗](#)), not only as a relay for emotional reactivity, but as a multifunctional hub embedded in complex social networks (Bickart et al., 2014 [↗](#)). Such complex patterns of behavioural implications are also reflected at the cellular level, as amygdala is composed of several different nuclei that are broadly connected with other brain areas that may displayed opposite functions (Zikopoulos et al, 2017;

Giacometti et al., 2023). Rather than opposing social cognition and emotion, our results support the view that emotional processing is deeply intertwined with social function—both being subserved by overlapping neural circuits (Domínguez-Borràs and Vuilleumier, 2022 [↗](#)). It also suggests that additional neural mechanisms, particularly those involving prefrontal and anterior cingulate regions implicated in the top-down regulation of affect and social behavior, may contribute to shaping species differences in social tolerance (Ochsner and Gross, 2008 [↗](#)). While our analysis compares social tolerance grades with variations in brain structure, the originality of our framework also lies in introducing three cognitive dimensions that bridge behavioral traits and neural substrates. This intermediate level of interpretation allows us to move beyond simple grade-to-structure associations, toward a more mechanistic understanding of the links between social behavior, cognition, and neuroanatomy.

Amygdala volume has also been shown to correlate positively with social network complexity in grade 1 species, as measured by the social network size of individuals (Sallet et al., 2011 [↗](#); Testard et al., 2022 [↗](#)), or by the social status of the animals (Noonan et al., 2014b [↗](#)). This supports the idea that the amygdala is sensitive to both structural social features and dynamic aspects of social networks.

Developmental trajectories and life-history plasticity

We are then led to question the origin of the social tolerance effect on amygdala volume, not in terms of a rigid nature versus nurture dichotomy, but in terms of differential developmental trajectories. Cross-fostering experiments (De Waal and Johanowicz, 1993 [↗](#)), along with our own results, suggest that social tolerance grades reflect both early, possibly innate predispositions and later environmental shaping. Moreover, the behavioral shifts observed in cross-fostered individuals underscore the plasticity of social style acquisition and the role of early social environment in shaping neural substrates of social behavior. Notably, tolerant species exhibit larger amygdala volumes early in life, while intolerant species show a progressive increase across the lifespan—a pattern that suggests a dual influence of biological programming and cumulative social experience. These environmental influences likely arise from both species-specific social dynamics—such as variations in affiliative behavior and social play (Beltrán Francés et al., 2020 [↗](#))—and broader ecological conditions that structure the demands of social life. The age-related volumetric changes we observed, particularly the divergence in developmental trajectories between tolerant and intolerant species, reinforce this idea and echo previous reports of amygdala growth patterns in humans and macaques (Schumann et al., 2019 [↗](#); Uematsu et al., 2012 [↗](#)). Taken together, these elements support the view that social tolerance is not fixed but emerges from the interplay between inherited developmental programs and the specific socio-ecological environments in which individuals mature.

Notably, the developmental trajectory of the amygdala in tolerant species does not align with that of intolerant species or with human developmental patterns (Schumann et al., 2019 [↗](#); Uematsu et al., 2012 [↗](#)). This finding suggests that neurodevelopmental pathways may exhibit significant variation among phylogenetically closely related primate species, potentially serving as an effective evolutionary target for adapting socio-ecological behaviours to environmental demands. Moreover, we observed that in old individuals (typically above 19 years), relative amygdala volume in grade 1 species could match that of grade 4 species — despite being significantly smaller earlier in life. Due to a limited sample size of our study, this crossing trend, already accounted for by our continuous age model, should be further investigated. These results call for cautious interpretation of age-related variation and further emphasize the importance of longitudinal studies integrating both behavioral, cognitive and anatomical data in non-human primates, which would help to better understand the link between social environment and brain development (Song et al., 2021 [↗](#)).

Hippocampal Volume and Social Cognitive Demands in Tolerant Species

A credible difference in hippocampal volume favoring SG4 individuals was only revealed between approximately 13 and 18 years of age by our hypothesis testing using a ROPE framework. Outside this range, the difference remained overall positive but inconclusive. This restricted window of significance, along with the unidirectional trend across the lifespan, suggests that increased hippocampal volume may nonetheless be associated with higher social tolerance, at least adulthood. At first glance, this observation may appear to contrast with literature linking chronic stress to reduced hippocampal size (Kim et al., 2015 [↗](#); Lyons et al., 2007 [↗](#)). However, as previously discussed, *M. tonkeana* (a high-tolerance species) combines elevated basal cortisol levels with a relatively large hippocampus (Sadoughi et al., 2021 [↗](#); Vandeleeuw et al., 2016 [↗](#)), which suggests that glucocorticoid exposure alone does not account for hippocampal variation in this context. Instead, our findings are more consistent with the idea that hippocampal structure reflects species-specific cognitive demands associated with navigating complex and tolerant social environments—such as spatial memory, social recognition, or contextual learning (Han et al., 2021 [↗](#); Sallet et al., 2011 [↗](#)). Within the conceptual framework introduced in this study, these results point to the importance of socio-cognitive requirements—rather than social environmental unpredictability or behavioral inhibition abilities—as potential drivers of interspecific variation in hippocampal anatomy. Comparative measurements and observations at individual level along with in vivo MRI from these same considered individuals may help to further understand how social tolerance can relate to cognitive abilities and its neural underpinning.

Limits of the study and future directions

While our dataset is comprehensive in terms of the number of macaque species included certain limitations must be acknowledged. For instance, phylogenetic analyses were beyond the reach of this study and integrating these statistical approaches could clarify the extent to which interspecific differences in brain structure and social behavior are due to shared ancestry or convergent evolution (Ghosh et al., 2022 [↗](#); Heuer et al., 2025 [↗](#)).

Although we explained some interspecies variability, adding subjects to our database will increase statistical power and will help addressing potential confounding factors such as age or sex in future studies. One will benefit from additional information about each subject. While considered in our modelling, the social living and husbandry conditions of the individuals in our dataset remain poorly documented. The living environment has been considered, and the size of social groups for certain individuals, particularly for individuals from the CdP, have been recorded. However, these social characteristics have not been determined for all individuals in the dataset. As previously stated, the social environment has a significant impact on the volumetry of certain regions. Furthermore, there is a lack of data regarding the hierarchy of the subjects under study and the stress they experience in accordance with their hierarchical rank and predictability of social outcomes position (McCowan et al., 2022 [↗](#)). In addition, our treatment of sex differences was limited. Although sex was included as a covariate in the Bayesian models, the strong imbalance in our dataset—favoring females (2:1 ratio)—precludes robust conclusions about sex-specific trajectories. Some trends, particularly regarding hippocampal volume, suggest potential interactions between sex, age, and social grade, but these effects remain exploratory. Addressing them adequately would require larger and more balanced samples, along with behavioral or hormonal data to capture intra-sexual variability. It is therefore important to recognize that confirmation of our findings should be achieved by analyzing datasets in which all of these confounding factors can be controlled more effectively.

While our study identifies the amygdala as a key subcortical structure associated with interspecific variation in social tolerance, it is important to acknowledge several neuroanatomical limitations. First, our analyses were conducted on the amygdala as a whole, without distinction between its internal nuclei. Although we used the SARM atlas (Hartig et al., 2021 [↗](#)), which offers a high-quality parcellation for *Macaca mulatta*, the precision of this template does not allow for

fully reliable automatic segmentation of amygdala subnuclei across the diverse species included in our dataset. As a result, our volumetric measures may conflate distinct functional subregions, potentially masking more localized effects. In this context, histological approaches remain essential for characterizing fine-grained neuroanatomical differences, as illustrated by Jones et al. (2021) [\[1\]](#), who reported interspecific variation in cell density and serotonergic innervation within the amygdala (Jones et al., 2021 [\[1\]](#)). Future studies combining MRI-based volumetry with *post-mortem* histology would allow more precise identification of which subregions underlie the observed differences in social tolerance.

Cognitive and neural perspectives on our understanding of social tolerance

Future directions linking behavior, cognition and neuroanatomy could deepen our understanding of the roots of social tolerance among macaque species. This could lead to a better operationalization of the concept that could be applied to a wider range of non-human primate species. From a neural perspective, studying the cortical regions associated with social tolerance represents a promising yet ambitious goal. In fact, there is a variability within primate species in cerebral organization (Amiez et al., 2023 [\[2\]](#), 2019 [\[3\]](#)) which is likely to be found across the *Macaca* genus. Considering this cerebral variability would require extensive efforts to properly assess interspecies differences, making it beyond the scope of the current study that focus on subcortical areas. However, as a starting point, exploring the connections between the amygdala, hippocampus, and medial prefrontal cortex could provide crucial insights into the neural correlates of social tolerance. These regions are central to stress regulation, socio-cognitive processing, and decision-making, all of which are likely impacted by social tolerance grades (Caetano et al., 2021 [\[4\]](#); Coplan et al., 2014 [\[5\]](#); Kim et al., 2015 [\[6\]](#); Phelps and LeDoux, 2005 [\[7\]](#); Sapolsky et al., 1990 [\[8\]](#)). In humans, repeated positive or stressful experiences have been demonstrated to alter the size of subcortical brain areas such as the hippocampus or amygdala (Davidson and McEwen, 2012 [\[9\]](#)) and impair neuroplasticity (Phelps, 2006 [\[10\]](#)). Neuronal plasticity and learning have been identified as contributing factors to variations in the ROI volume, including the amygdala and hippocampus, particularly in humans (Maguire et al., 2000 [\[11\]](#); Taren et al., 2013 [\[12\]](#)). Additionally, our conceptual framework opens avenues for advanced neuroimaging techniques such as diffusion tensor imaging (DTI) (Howard et al., 2023 [\[13\]](#); Zhang et al., 2013 [\[14\]](#)) or multiparametric MRI (Mulholland et al., 2024 [\[15\]](#)), which could be used to explore white matter connectivity or microstructural changes. Our findings also emphasize the need to develop individual-level measures of social tolerance (Dubuc et al., 2012 [\[16\]](#); DeTroy et al., 2021 [\[17\]](#)). Fine-tuning these measures would allow more precise correlations between behavioral data and neuroanatomical features. By operationalizing the concept of social tolerance on cognitive dimensions, our work aims at enriching the framework through which primate sociality is currently studied.

Conclusion

Our study provides novel insights into the relationship between amygdala volume and social tolerance in macaques, offering an innovative perspective on the neuroanatomical basis of social cognition. Using a comparative approach across 12 macaque species, we uncovered a revealing relationship: low-tolerance species start their life with a smaller amygdala compared to their socially tolerant counterparts. In addition, intolerant species show an increase in amygdala volume, whereas highly tolerant species show the opposite trend. These findings refine conventional views of the amygdala by highlighting its broader role in both emotional regulation and complex social cognition. The observed differences in amygdala volume with respect to social tolerance grades suggest that the development and plasticity of the amygdala seem to be intricately linked to the social environment and experiences of the species. Larger amygdala in socially tolerant species may reflect an enhanced capacity to process complex social information, facilitating better social interactions, cooperative behavior, and conflict management. Alternatively, the observed increase in amygdala volume in socially intolerant species over time

may be explained by heightened socio-cognitive demands, rather than being solely attributed to chronic stress or emotional reactivity. While earlier studies emphasized the role of the amygdala in stress response, recent findings are in lines with our results which suggest that amygdala's functions extend to broader aspects of social cognition. These findings have profound implications for our understanding of social brain evolution as well as underscoring the importance of developmental stage and the social environment being crucial drivers of neuroanatomical adaptations. In addition, although hippocampal volume showed less pronounced and more variable differences across social grades, a credible effect was observed between 13 and 18 years of age. Across all ages, SG4 individuals consistently exhibited larger hippocampal volumes than SG1, supporting the possibility that this region also contributes to social cognitive processes in tolerant species—especially during developmental phases associated with social maturation. This study, at the interface of primatology and cognitive neuroscience, also provides a framework for investigating the impact of the social environment on brain development and pave the way for future research to unravel the complexities of brain evolution and sociality.

Material & Methods

Brain specimen collection

To allow comprehensive cross-species comparisons in the *Macaca* genus, a dataset of 43 *post-mortem* specimens has been constituted through collaborations with multiple research centers, each contributing unique expertise and resources (Table 2 [↗](#)). The collaborating institutions included (Table S1 [↗](#)):

The **Centre de Primatologie de l'Université de Strasbourg (CdP)**: provided valuable brain data derived from 20 brain samples. Among those, one sample (*M. nigra*) was obtained as part of a collaboration with the zoo of Mulhouse (<https://www.zoo-mulhouse.com/↗>).

Samples from INDI-PRIME-DE (Milham et al., 2018 [↗](#)): **The Japan Monkey Center** provided 5 *post-mortem* MRI acquisitions to the dataset (Sakai et al., 2018 [↗](#)). **Utrecht University**: contributed to the dataset with 13 *post-mortem* MRI acquisitions (Navarrete et al., 2018 [↗](#)).

INSERM-Oxford University: 5 *post-mortem* MRI acquisitions came from this collaboration. This addition offered more variety of acquired data mostly in age and sex (Milham et al., 2018 [↗](#)).

Ethical considerations

The study was conducted in accordance with ethical guidelines and was approved by the ethical committee of the Centre de Primatologie de l'Université de Strasbourg which is authorized to house NHP (registration B6732636). The research further complied with the EU Directive 2010/63/EU for animal experiments. All subjects from the CdP died of natural or accidental cause; no macaque was euthanized in the sole frame of the project. These specimens originated from CdP, and their collection followed rigorous ethical considerations. The specimens were either obtained from previous collections—where full bodies were preserved in dedicated freezers—or from individuals of the CdP that had died from natural causes. The *post-mortem* MRI data from INSERM-Oxford University were acquired from deceased animals that died of causes unrelated to the present research project. As such, the research did not require a Home Office License as defined by the Animals (Scientific Procedures) Act 1986 of the United Kingdom.

Brain extraction technique

Post-mortem MRI images acquisition of macaque brains is central to our study, more specifically, in translational studies of homologous brain regions. Brain extraction is a crucial process in neuroscience research for studying the internal brain structure of animals. Through the acquisition at the CdP of 20 *post-mortem* anatomical MRI scans of brains from six different species of macaques, we were able to refine a brain extraction technique - whether previously frozen or fresh - to minimize specimen handling artefacts and obtain image quality suitable for optimal use by the scientific community. The detailed extraction technique protocol established and used for

our brain extractions is available in **Document S1**. Briefly, the head is reclined forward to expose the neck, muscles are removed to access the atlanto-occipital junction, which is then incised to allow head dislocation (Figure S1 [↗](#)). An osteotome and hammer are used, ensuring no cerebellar herniation. The skull cap is carefully drilled using a rotary tool and removed (see Figure S1. A, B [↗](#) and Table S2 [↗](#) for required tools), and the brain is extracted by severing the olfactory peduncles, internal carotid arteries, and cranial nerves. Specimens are then fixed in 10% buffered formaldehyde for 7 days (see Figure S1. D. [↗](#)) and in phosphate buffered saline (PBS) for 3–4 days before being placed in Fluorinert™ for MRI acquisition, ensuring minimal air bubbles and optimal image quality (Sébille et al., 2019 [↗](#)) see Figure S2 [↗](#).

Sampling methods & measurements

Structural images were collected through both the open access databases and collaborations (Milham et al., 2018 [↗](#); Navarrete et al., 2018 [↗](#); Sakai et al., 2018 [↗](#)), but also carried out at the IRIS platform of the ICube laboratory in Strasbourg for *post-mortem* samples kept at the CdP (see Table S3 [↗](#) for the information relating to the acquisition of anatomical MRI images). The final dataset consists of 43 anatomical scans after pruning data with missing age or sex information (10 individuals), with both T₁ and T₂-weighted images. Due to their different origins, the images in the dataset did not follow the exact same acquisition protocols (different scanners and acquisition parameters, Table 2 [↗](#)). In addition, *post-mortem* brain preservation and perfusion protocols are different, which may also influence the images obtained. Volume measurements were performed using a semi-automatic method to register individual images to the Subcortical Atlas of the Rhesus Macaque (SARM) (Hartig et al., 2021 [↗](#)) (Figure S1.E. [↗](#)). Due to the large orientation discrepancies across the research centers, the images were first manually realigned (translation and rotation) with the atlas using ITK-SNAP (Yushkevich et al., 2006 [↗](#)), then non-linearly registered using ANTs (Avants et al., 2011 [↗](#)). The segmentation maps of the atlas were then transported to the subject space to extract the volume of the regions of interest. Figure S1 [↗](#) details the image processing for volume extractions (see Figure S2 [↗](#)). To ensure the accuracy of the SARM on our dataset, which includes 11 species other than *M. mulatta* (the species used for SARM development (Hartig et al., 2021 [↗](#))), we calculated the Dice Similarity Coefficient (DSC) (Zou et al., 2004 [↗](#)). This was done by manually segmenting, using a tablet (Wacom Cintiq® 16 and ITK SNAP software), the amygdala in each acquisition and comparing the overlapping voxels between the manual segmentation and the SARM segmentation. With a DSC of 0.96, we confirm the robust performance of the SARM across our entire dataset.

Final dataset characteristics

The dataset is composed of 12 distinct macaque species with a total of 43 individual specimens for analysis. There is a strong sex imbalance with more females than males. The age range spans from 1 to 44.20 years, with an average age of 18.2 ± 9.4 years (standard deviation) (Figure 5. B, D.) [↗](#) with two outliers above 35 years old. Most importantly, based on our research question, the social grade distribution of our dataset (Figure 5. A.) [↗](#) is more represented by grade 1 than grade 4 species as these species are very rare in zoo or in research centers and most of them protected as endangered species. The MRI acquisitions from the 43 individuals were standardized to the NMT template (Jung et al., 2021 [↗](#)). Data included amygdala or hippocampus volume and a computed brain “total volume” which only excludes the myelencephalon and the cerebellum for reliability. In fact, the integrity of these subcortical structures heavily depends on the quality and technics used for brain extraction methods.

Modelling approach

To investigate the subcortical correlates of social tolerance in macaques, we used a multivariate Bayesian linear model with normal likelihood, the observed data being the amygdala and hippocampus volume. The predictors in our model were the intercept, social grade, age, sex, husbandry, whether the brain had been previously frozen, total volume, the interaction between social grade and age, and the covariance between the observations. We used wide priors, whose

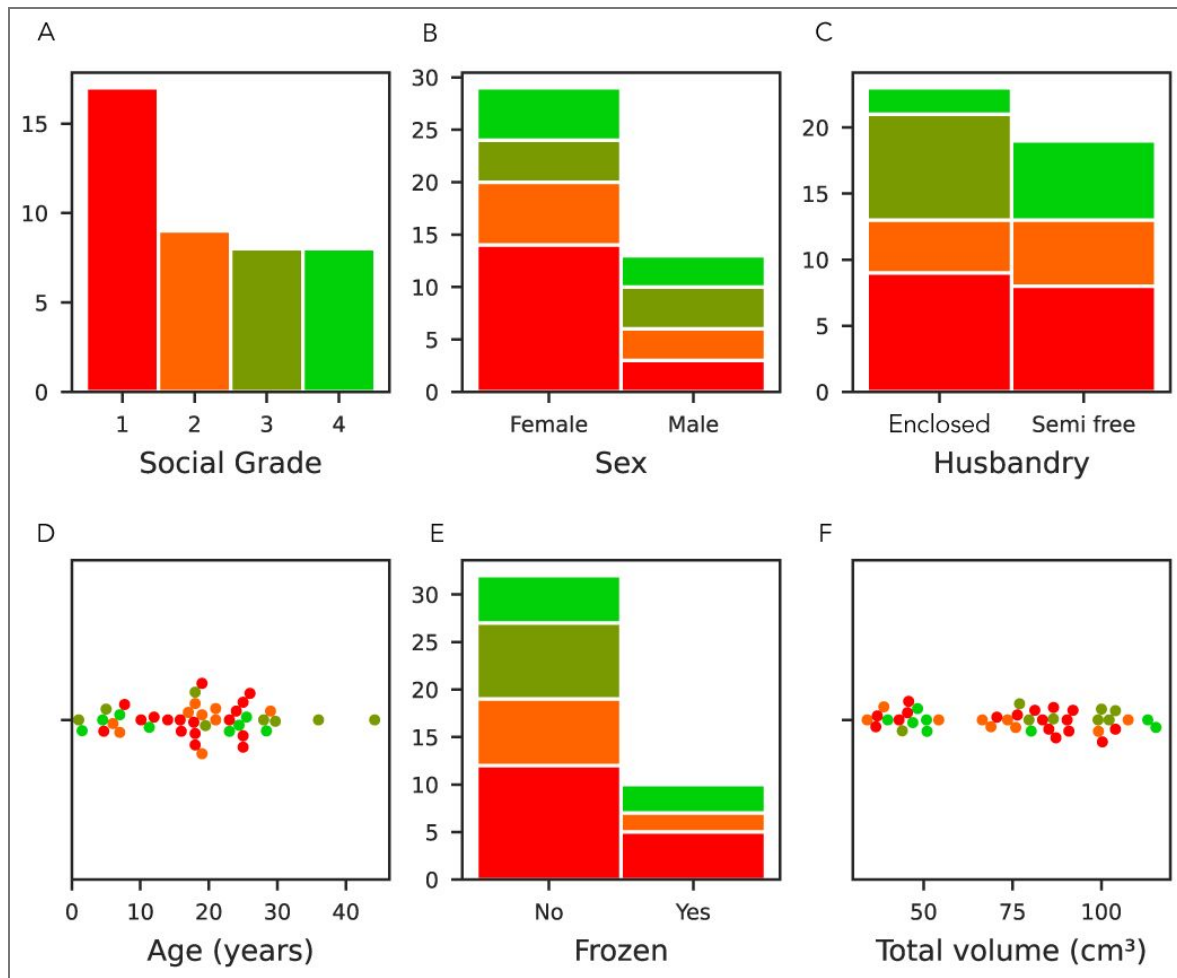


Figure 5. Dataset characteristics relative to the social grade.

In red: social tolerance grade 1, orange: grade 2, olive: grade 3, and green: grade 4. **(A)** Social tolerance grade distribution, where grade 1 is overrepresented due to the prevalence of *Macaca mulatta* in laboratories. **(B)** Sex distribution: There was a significant imbalance in the sample, with females outnumbering males (2:1 ratio). **(C)** Husbandry distribution of the individuals (enclosed and semi-free ranging conditions) **(D)** Age distribution: The cohort had a relatively even age distribution with a notable peak in the 20s. **(E)** The frozen status distribution. **(F)** Total brain volume distribution, excluding the myelencephalon and cerebellum due to variation in their preservation.

locations and scales were derived from the data. We assessed the quality of the model by comparing the predicted data to the observed data, and by checking the R^2 of the model. New data was predicted to study the interaction and the age-social grade trajectory, and the difference in volume between social grades. The predictions were made using the model on all social grades, on females aged from 1 to 40, with a total volume of 85 cm³.

Supplementary

The dataset includes brain imaging data from 42 individuals across 12 macaque species. Columns represent the collection centers: CdP (Centre de Primatologie de l'Université de Strasbourg, France), Utrecht University (Netherlands), INSERM-Oxford University (Fr/UK), and Japan Monkey Center (Japan). Each row corresponds to a macaque species, with the number of individuals collected from each center. The total count per species and per center is indicated in the final column and row, respectively. The social grade of the species is indicated between brackets.

Document S1: Detailed brain extraction procedure

The first part of the acquisition of brain MRI images relies on the extraction of brain specimens. All the required materials are listed in [Table S1](#) and have been curated through literature recommendations ([Brown et al., 2009](#); [Davenport et al., 2014](#); [King et al., 2013](#); [Shatil et al., 2016](#)) as well as through the realization of the procedure itself.

As it was the first time at the CdP that brain specimens were extracted, the development of a refined brain extraction technique played a central role in optimizing the quality of the acquired brain images. This technique, based on the literature, was meticulously applied to 17 *post-mortem* anatomical and DTI (if the specimen was not frozen) MRI scans. We optimally prepared through familiarizing with the technique and handling instruments such as the oscillating saw with a first brain extraction of a poor conditioned brain specimen (bad freezing condition). Through this first trial, we were able to adjust the following procedure as well as the chosen cutting landmarks. The brain extraction sequence consists of the following steps:

Head Dislocation: The animal is in supine position, with its head pulled dorsally to extend the neck ([Brown et al., 2009](#)). Then, the muscle mass of the neck has to be removed to expose the atlantooccipital junction. The junction is then incised, employing a back-and-forth motion to enable the passage of a knife or hammer-axis between the cartilage ([Brown et al., 2009](#)).

Adjustment for Frozen Specimens: For frozen specimens, a distinct approach through the atlantooccipital junction must be adopted. The section is performed with an osteotome and hammer ([Figure S1, A](#)). Subsequently, confirmation of the absence of cerebellar protrusion through the foramen magnum is required and essential for ensuring the integrity of the specimens. If not, it would indicate cerebellar herniation and the presence of associated lesions in this brain structure ([Davenport et al., 2014](#)).

The head circumference was measured using a measuring tape during data collection. The average circumference of a macaque brain in our dataset is 30.5 ± 8.7 cm. We determined the amount of formaldehyde required based on the average volume of a macaque brain. This average volume is of 89.2 ± 1.9 (SEM) for male individuals and of 70.8 ± 0.72 cm³ for female individuals ([Franklin et al., 2000](#); [Scott et al., 2016](#)). The amount of 4 % formaldehyde buffered solution (pH = 6.9, Sigma-Aldrich) required to fix the brain should follow the volume ratio of tissue to be fixed to formaldehyde of 1:10 ([Thavarajah et al., 2012](#)), that is, a minimum of one liter of formaldehyde.

Skull Cap Removal: The removal of the skull cap is the most meticulous part of the procedure aimed at providing access to the brain. This process starts with a longitudinal incision of the scalp from the anterior fontanel cranial suture to the foramen magnum. A second incision is made perpendicular to the first, along the coronal suture. The exposed skull surface underwent thorough cleaning with 70% ethanol, followed by drying with gauze pads. The bony skull cap was then delicately excised using a rotary tool ([Figure S1, B](#)).

Figure S1. Total brain volume across macaque species categorized by social grade.

Distribution of total brain volumes (in cm³) across the four social tolerance grades of the *Macaca* genus. Each dot represents an individual (n=42), and colors indicate social grade: red (grade 1, intolerant), orange (grade 2), olive (grade 3), and green (grade 4, tolerant). Total brain volume was computed from *post-mortem* MRI scans, excluding the cerebellum and myelencephalon to control for inter-individual variation in preservation quality. While total volume was included as a covariate in the statistical model, this figure provides a complementary descriptive overview of its distribution across social grades.

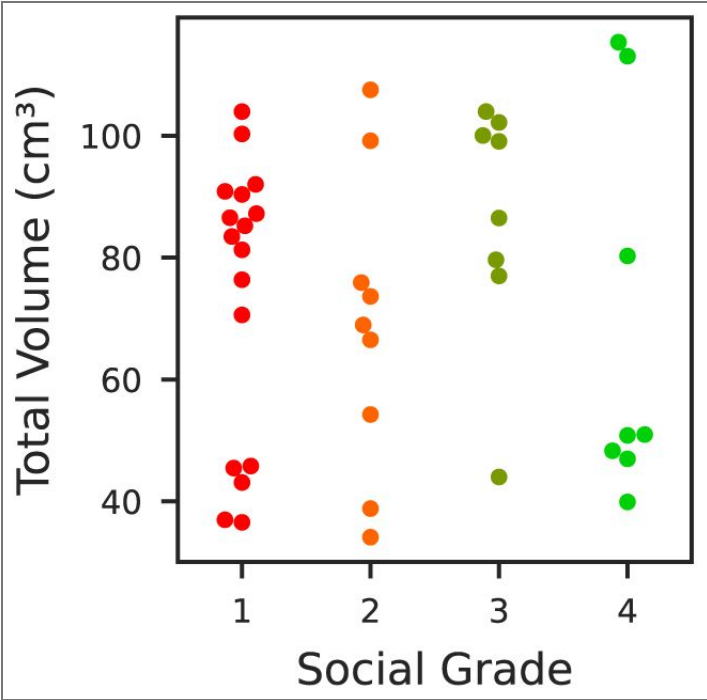


Table S1. Species and data collection centers in the dataset

	CdP	Utrecht University	Inserm Oxford University	Japan Center	Monkey	Total
<i>M. arctoides</i> [3]		3				3
<i>M. fascicularis</i> [2]	3	1	1	1		6
<i>M. fuscata</i> [1]		2		2		4
<i>M. mulatta</i> [1]	8	1	4			13
<i>M. nemestrina</i> [2]	1	1				2
<i>M. nigra</i> [4]	1	1				2
<i>M. silenus</i> [3]		1				1
<i>M. sinica</i> [3]				1		1
<i>M. sylvanus</i> [3]		2				2
<i>M. thibetana</i> [2]	1					1
<i>M. tonkeana</i> [4]	6					6
<i>M. radiata</i> [3]				1		1
Total	20	12	5	5		42

Summary table of our dataset detailing the macaque species, the social grades [x] and the centers they originate from.

Materials	Description
Multifunctional rotary tool (Dremel type) associated with a sawing disc	To precisely cut out and remove the skull and have access to the brain.
Osteotome	To separate the skull from the atlas.
Mallet	Used with the osteotome.
Hammer-axe	To separate the skull from the atlas and to enlarge the space between the muscular planes and the skull.
Autopsy knife	To cut out the flesh and muscles.
Scalpel handle and blade	To dissect the scalp, muscles of the head, the dura, and the facial nerves.
Grooved probe	To dissect the dura without damaging the underlying tissues.
Pair of curved scissors with round tip	To cut out flesh at various steps of the dissection.
Dissecting forceps with and without claws	To hold tissues without damaging them to perform sections (skin, muscles, dura, nerves).
Graded beakers	To measure and transfer formaldehyde into the bucket for the brain fixation.
Funnel	To transfer the various liquids without spilling.
Test tube with graduation	To measure out the required amount of distilled water for the PBS.
Buckets (x4)	Containers for the brain fixation.
Hemispheric receptacles	With holes at regular intervals, to hold the brain spherical shape whilst it is immersed in formaldehyde.
Plastic jar with screw lid	Transparent, 7 centimeters in diameter (adapted to the size of the MRI antenna used). Container for brain during MRI acquisitions.
Fluorinert™ FC-770	Fluorocarbon-based fluid, helps to displace air bubbles ^[1] and improves brain/background image delineation.

Table S2. Necessary materials for NHP brain extraction– name and description.

Distilled water	To dissolve the PBS tabs to make PBS solution.
Parafilm	To seal the MRI containers and prevent any leakage.
Cotton batting/aquarium foam squares	To place inside the MRI container at the bottom, the top and on one side. Used to pad the brain specimen and contain the bubble at the top of the container.
FFP2 mask	Safety measures for zoonosis.
Latex or nitril surgical gloves	For handling specimen brains.
Lab coats (or surgical gowns)	For individual protection whilst performing the dissection.
Marker	To label the various containers (mentioning if it has contained formaldehyde), label the brain specimen.
7T MRI Antenna	For the MRI acquisition (whole brain acquisition). Check the internal diameter to choose the adequate container.
Material for the use of paraformaldehyde	
Formaldehyde solution 4 %	pH 6,9 solution for histological tissue fixation. Ordered from Sigma-Aldrich (Product ID: HT501640-19L.) At least 10 x volume of a macaque brain (roughly 3 liters per brain).
Formaldehyde spill response kits	Used for safety in case of any formaldehyde spilling.
Fume hood	To manipulate formaldehyde ensuring the safety of the experimenter.
Face shield	For the experimenter's safety in case of any spilling.
Mask	For the experimenter's safety in case of any spilling.
Gloves (chemical resistant)	For the experimenter's safety for handling the formaldehyde.
Baritainer	For discarding waste

^[1] (Iglesias et al., 2018)

Summary table of the tools required for the brain extraction we performed, including their names and descriptions.

Table S2. (continued)

Brain Extraction: The extraction of the brain from the skull was conducted with care especially for fresh specimens as the tissues are very soft and breakable. Removal of the dura mater, cerebellum tent, and false brain is performed using a bony cap (Brown et al., 2009 [↗](#)). Afterwards, the head is positioned vertically and some gentle taps on the table to facilitate the gradual detachment of the brain from the skull. Employing a fluted probe, the olfactory peduncles, internal carotid arteries, and cranial nerves are delicately severed (King et al., 2013 [↗](#)). The pineal gland is commonly found as a single white firm conical mass suspended at the midline of the skull cap and attached to the meninges (King et al., 2013 [↗](#)). In addition, in many animals, the choroid plexus is visible as two reddish masses in a position similar to or slightly in front of the pineal gland (King et al., 2013 [↗](#)). Once the brain is fully extracted, it is placed in one of the hemispheric cups. Frozen specimens are thawed in water at room temperature before placing them in the cup (frozen tissue combined with formaldehyde fixation creates an aqueous insulating layer, altering fixation of the internal brain structures) (Figure S1. D [↗](#)).

Fixation and Brain Preparation for MRI

The fixation and preparation of brain specimens for MRI acquisitions is a crucial phase to ensure high quality images and low prevalence of artefacts. The chosen fixative was a 10 % formaldehyde buffered solution. Ensuring complete immersion of the brain in the fixative is crucial. To achieve this, a carefully calculated volume of the solution is added to a labelled container (usually around 3L). The brain is then placed within the container, submerged to guarantee full coverage. Sealing the container prevents contamination. The container remains for 7 days under the fume hood, and we check and gently stir every day to ensure the good repartition of the formaldehyde on the tissue. Once the brain is fixed, it is immersed the brain in a 0.5 L jar filled with PBS for 48 hours before MRI.

1. MRI acquisitions

The diameter of the plastic container was chosen based on the diameter of the MRI antenna used (8,6 cm in diameter).

- The sample is placed in a spherical container. The orientation in which the brain is placed (for future reference during the MRI acquisitions) was registered. Aquarium foam squares are placed around the brain to minimize the residual movements from the MRI's vibrations and to contain the remaining air bubble at the top of the container (Figure S2. B [↗](#)).
- The container is filled to the brim with Fluorinert™ FC-770, a liquid that optimizes the contrast of the MRI signal and allows the wobble adjustment.
- The container is placed in a vacuum chamber (negative pressure of $-0,1$ Pa.), to limit the presence of air bubbles on the images, for up to three hours to remove air bubbles (Shatil et al., 2016 [↗](#)) (Figure 1, A [↗](#)). If needed, some Fluorinert™ FC-770 can be added up to the brim.
- The container is sealed with parafilm (Figure S2. B [↗](#)).

2. Recommended 7T-MRI scanning setup

Place the container with an elevating foam square to contain the remaining bubble at the top of the jar and limit the superimposition of the bubble on the brain (Figure S2. C and D [↗](#)).

Once in the MRI, it is recommended to perform a sequence of localizer scans with the purpose of: (1) identifying significant distortions caused by air bubbles in the brain or MRI-compatible container, (2) accurately positioning the brain, and (3) establishing the slice positions necessary for subsequent data acquisition (Shatil et al., 2016 [↗](#)).

Data availability

The data associated with this study are available at DOI:
<https://doi.org/10.17605/OSF.IO/AQMSW> [↗](#).

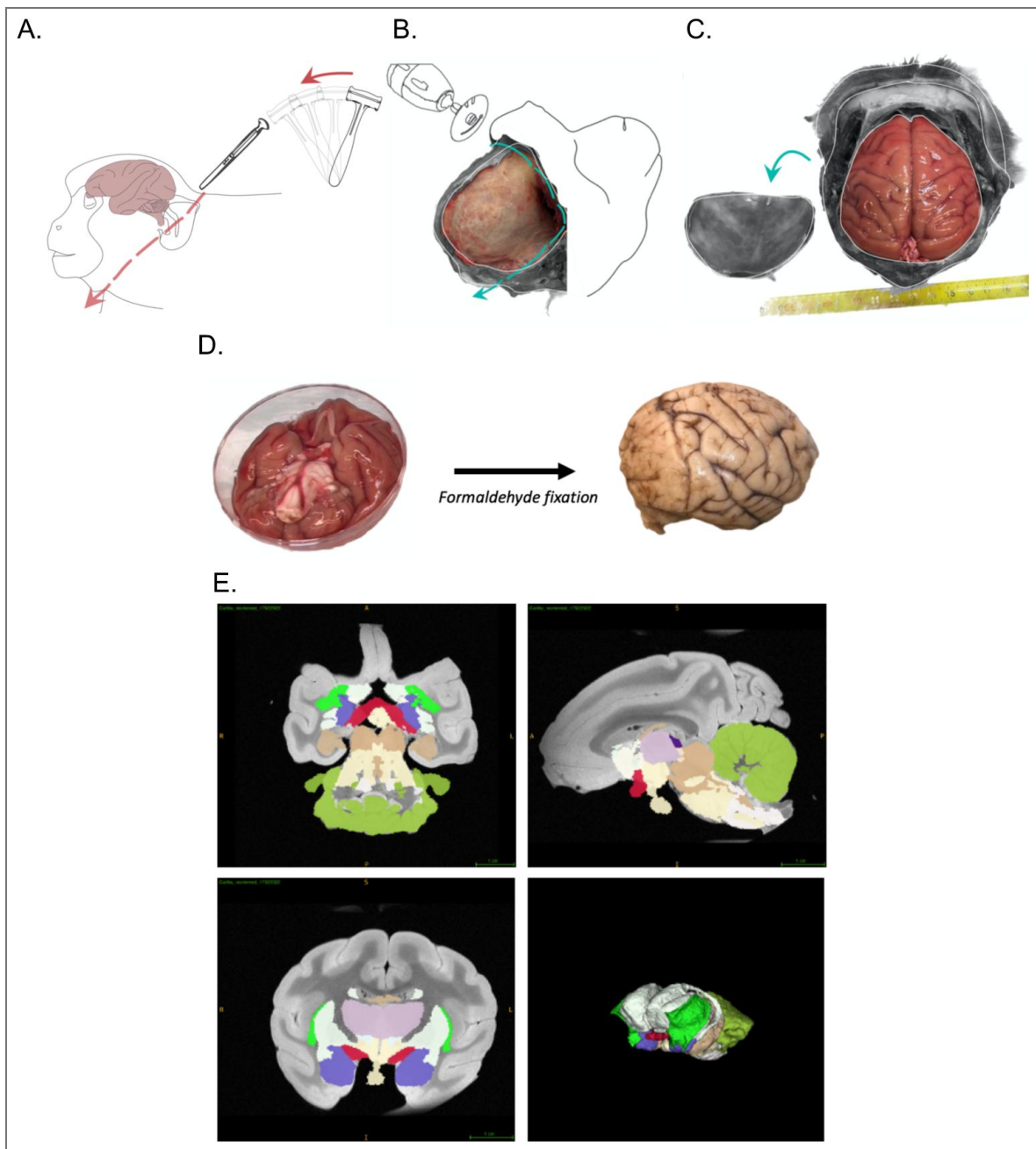


Figure S1. Sequence of dissection steps and MRI acquisition.

(A) Craniotomy step: Position of the cadaver and cut site. (B) Steps of scalp and skull removal using a Dremel tool[®] associated with a flex shaft rotary tool. (C) View of the skull after skull and dura removal. (D) Extraction and formaldehyde fixation of the brain. Right lateral view of the brain after a 7 day-formaldehyde-fixation. (E) SARM Regions (SARM2) and 3D MRI acquisition with atlas (bottom right). Amygdala (purple) (Hartig et al., 2021 [DOI](#)).

Figure S2. Set of photographs of the preparation of the fixed brain for MRI acquisitions.

(A) Air bubble removal stages in a vacuum chamber. The brain is immersed in Fluorinert™ FC-770 and held in position by the aquarium foam squares. The container is placed in a receptacle to catch any Fluorinert™ FC-770 that may spill out of the container during the procedure; (B) Placement of the aquarium foam squares inside the container of brain immersed in Fluorinert™ FC-770 and sealed with parafilm; (C) Placement of the container with a lift foam square to contain the residual air bubble in the top third of the container; (D) Placement of the container in the MRI antenna.

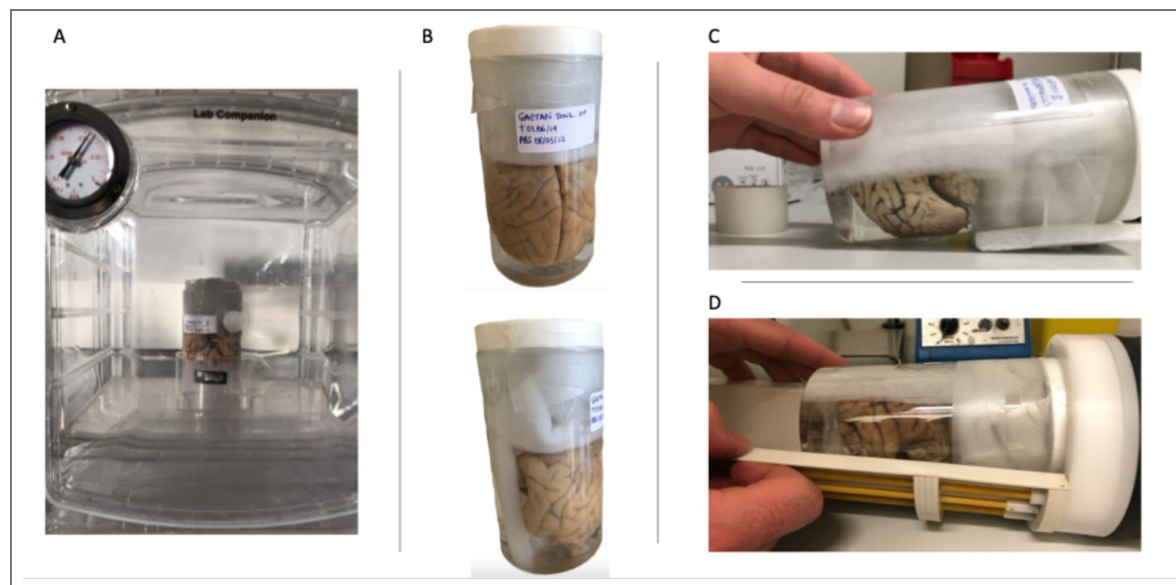


Table S3. Information relating to the acquisition of anatomical MRI images and the procedures for fixing and preserving post-mortem samples according to the different centers.

MRI Center	Primate Center (with laboratory)	Utrecht University	Japan Monkey Center	Oxford University
Number of individuals	19	14	4	4
MRI	Bruker Biospec 7T 70/30 UR	Agilent Variant 9.4T	Bruker Biospec 9.4T 90/20 UR	Clinical Horizontal 3T
Antenna	Volume coil	Surface coil	Volume coil	Volume coil
MRI weighting	T2 Sequence: Turbo_RARE_3D	T1	T2	T2
Spatial resolution	0.20 x 0.20 x 0.20 mm	0.25 x 0.25 x 0.25 mm	0.25 x 0.25 x 0.25 mm	0.6 x 0.6 x 0.6 mm
Brain fixation characteristics	Immersion: Formaldehyde 4%	Perfusion: Formaldehyde 4%	Immersion: formaline 10%	Perfusion: Formaldehyde 4%
Fluid used for scanning	Fluorinert® FC-770	Sucrose 15%, 4°C	3M™ Fluorinert	Fomblin or Fluorinert® FC-770

Summary table of the MRI acquisitions parameters and *post-mortem* sample preservation protocols across centers.

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Additional information

Author’s contribution

S.B. and S.S. conceptualized the research; C.P., S.B., J.S. and S.S. developed the methodology, S.S. performed the data curation, J.L. performed the formal analysis, S.S., S.B. C.P., J.S. and M.L. performed the investigation; S.S., J.L. and S.B. wrote the original manuscript; S.B. and J.S. acquired the funding. All the authors reviewed the manuscript.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This paper investigates the potential link between amygdala volume and social tolerance in multiple macaque species. Through a comparative lens, the authors considered tolerance grade, species, age, sex, and other factors that may contribute to differing brain volumes. They found that amygdala, but not hippocampal, volume differed across tolerance grades such that high-tolerance species showed larger amygdala than low-tolerance species of macaques. They also found that less tolerant species exhibited increases in amygdala volume with age, while more tolerant species showed the opposite. Given their wide range of species with varied biological and ecological factors, the authors' findings provide new, important evidence for changes in amygdala volume in relation to social tolerance grades. Contributions from these findings will greatly benefit future efforts in the field to characterize brain regions critical for social and emotional processing across species.

(1) This study demonstrates a concerted and impressive effort to comparatively examine neuroanatomical contributions to sociality in monkeys. The authors impressively collected samples from 12 macaque species with multiple datapoints across species age, sex, and ecological factors. Species from all four social tolerance grades were present. Further, the age range of the animals is noteworthy, particularly the inclusion of individuals over 20 years old.

(2) This work is the first to report neuroanatomical correlates of social tolerance grade in macaques in one coherent study. Given the prevalence of macaques as a model of social neuroscience, considerations of how socio-cognitive demands are impacted by the amygdala are highly important. The authors' findings will certainly inform future studies on this topic.

(3) The methodology and supplemental figures for acquiring brain MRI images are nicely detailed. Clear information on these parameters is crucial for future comparative interpretations of sociality and brain volume, and the authors do an excellent job of describing this process in full.

(4) The following comments were brought up during the review. In their revision, the authors have sufficiently addressed all of these comments by providing detailed responses and updating their manuscript. First, the revision clarified how much one could draw conclusions about "nature vs. nurture" from this study. Second, the revision also clarified the contributions of very young and very old animals in their correlations. Third, in their revision, the authors expanded on how their results could be interpreted in the context of multiple behavioral traits by Thierry (2021) by providing more detailed descriptions. Finally, during the revision, the authors clarified that both intolerant and tolerant species experience complex socio-cognitive demands and highlighted that socio-cognitive challenges arise across the tolerance spectrum under different behavioral demands.

<https://doi.org/10.7554/eLife.106424.2.sa3>

Reviewer #2 (Public review):

Summary:

This comparative study of macaque species and type of social interaction is both ambitious and inevitably comes with a lot of caveats. The overall conclusion is that more intolerant species have a larger amygdala. There are also opposing development profiles regarding amygdala volume depending on whether it is a tolerant or intolerant species.

To achieve any sort of power they have combined data from 4 centres - that have all used different scanning methods and there are some resolution differences. The authors have also had to group species into 4 classifications - again to assist with any generalisations and power. They have focussed on the volumes of two structures, the amygdala and the hippocampus, which seems appropriate. Neither structure is homogeneous and so it may well be that a targeted focus on specific nuclei or subfields would help (the authors may well do this next) - but as the variables would only increase further along with the number of potential comparisons, alongside small group numbers, it seems only prudent to treat these findings as preliminary. That said, it is highly unlikely that large numbers of macaque brains will become available in the near future.

This introduction is by way of saying that the study achieves what it sets out to do, but there are many reasons to see this study as preliminary. The main message seems to be twofold: 1) that more intolerant species have relatively larger amygdalae, and 2) that with development there is an opposite pattern of volume change (increasing with age in intolerant sp and decreasing with age in tolerant species). Finding 1 is the opposite of that predicted in Table 1 - this is fine, but it should be made clearer in the Discussion that this is the case otherwise the reader may feel confused. As I read it, the authors have switched their prediction in the Discussion, which feels uncomfortable.

It is inevitable that the data in a study of this complexity are all too prone to post hoc considerations, to which the authors indulge. I suspect I would end up doing the same but it feels a bit like 'heads I win, tails you lose'. In the case of Grade 1 species, the individuals have a lot to learn especially if they are not top of the hierarchy, but at the same time there are fewer individuals in the troop, making predictions very tricky. As noted above, I am concerned by the seemingly opposite predictions in Table 1 and those in the Discussion regarding tolerance and amygdala volume. (It may be that the predictions in Table 1 are the opposite to how I read them, in which case the Table and preceding text needs to align.)

Comments on revisions:

I am happy with all of the revisions and the care shown by the authors.

<https://doi.org/10.7554/eLife.106424.2.sa2>

Reviewer #3 (Public review):

Summary:

In this study, the authors were looking at neurocorrelates of behavioural differences within the genus *Macaca*. To do so, they engaged in real-world dissection of dead animals (unconnected to the present study) coming from a range of different institutions. They subsequently compare different brain areas, here the amygdala and the hippocampus, across species. Crucially, these species have been sorted according to different levels of social tolerance grades (from 1 to 4). 12 species are represented across 42 individuals. The sampling process has weaknesses ("only half" of the species contained by the genus, and *Macaca mulatta*, the rhesus macaque, representing 13 of the total number of individuals), but also strengths (the species are decently well represented across the 4 grades) for the given purpose and for the amount of work required here. I will not judge the dissection process as I am not a neuroanatomist, and I will assume that the different interventions do not alter volume in any significant ways / or that the different conditions in which the bodies were kept led to the documented differences across species.

There are two main results of the study. First, in line with their predictions, the authors find that more tolerant macaque species have larger amygdala, compared to the hippocampus

that remains undifferentiated across species. Second, they also identify developmental effects, although with different trends: in tolerant species, the amygdala relative volume decreases across the lifespan, while in intolerant species, the contrary occurs. The modifications brought up between the two versions of the article have answered my remarks regarding age/grade/brain area differences.

As such, I think the results are holding strong, but maybe more work is needed with respect to interpretation.

Classification of the social grade, as well as the issue of nature vs nurture have been addressed by the authors, I thank them for this.

I still feel the integration of the amygdala as a common cognitive & emotional center could be possibly more pushed in the discussion, although I acknowledge that it would be complicated to do without knowing how the emotional and social lives of these animals impacted the growth of their amygdala...

Strengths:

Methods & breadth of species tested

Weaknesses:

Interpretations, which, although softened, could still be more integrated with the literature on emotion

<https://doi.org/10.7554/eLife.106424.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

We thank Reviewer #1 for its thoughtful and constructive feedback. We found the suggestions particularly helpful in refining the conceptual framework and clarifying key aspects of our interpretations.

Summary:

This paper investigates the potential link between amygdala volume and social tolerance in multiple macaque species. Through a comparative lens, the authors considered tolerance grade, species, age, sex, and other factors that may contribute to differing brain volumes. They found that amygdala, but not hippocampal, volume differed across tolerance grades, such that hightolerance species showed larger amygdala than low-tolerance species of macaques. They also found that less tolerant species exhibited increases in amygdala volume with age, while more tolerant species showed the opposite. Given their wide range of species with varied biological and ecological factors, the authors' findings provide new evidence for changes in amygdala volume in relation to social tolerance grades. Contributions from these findings will greatly benefit future efforts in the field to characterize brain regions critical for social and emotional processing across species.

Strengths:

(1) This study demonstrates a concerted and impressive effort to comparatively examine neuroanatomical contributions to sociality in monkeys. The authors impressively collected samples from 12 macaque species with multiple datapoints across species age, sex, and ecological factors. Species from all four social tolerance grades were present. Further, the age range of the animals is noteworthy, particularly the inclusion of individuals over 20 years old - an age that is rare in the wild but more common in captive settings.

(2) This work is the first to report neuroanatomical correlates of social tolerance grade in macaques in one coherent study. Given the prevalence of macaques as a model of social neuroscience, considerations of how socio-cognitive demands are impacted by the amygdala are highly important. The authors' findings will certainly inform future studies on this topic.

(3) The methodology and supplemental figures for acquiring brain MRI images are well detailed. Clear information on these parameters is crucial for future comparative interpretations of sociality and brain volume, and the authors do an excellent job of describing this process in full.

Weaknesses:

(1) The nature vs. nurture distinction is an important one, but it may be difficult to draw conclusions about "nature" in this case, given that only two data points (from grades 3 and 4) come from animals under one year of age (Method Figure 1D). Most brains were collected after substantial social exposure-typically post age 1 or 1.5-so the data may better reflect developmental changes due to early life experience rather than innate wiring. It might be helpful to frame the findings more clearly in terms of how early experiences shape development over time, rather than as a nature vs. nurture dichotomy.

We agree with the reviewer that presenting our findings through a strict nature vs. nurture dichotomy was potentially misleading. We have revised the introduction and the discussion (e.g. lines 85-95 and 363-365) to clarify that we examined how neurodevelopmental trajectories differ across social grades with the caveat of related to the absence of very young individuals in our samples. We now explicitly mention that our results may reflect both early species-typical biases and experience-dependent maturation.

We positioned our study on social tolerance in a comparative neuroscience framework and introduced a tentative working model that articulates behavioral traits, cognitive dimensions, and their potential subcortical neural substrates

Drawing upon 18 behavioral traits identified in Thierry's comparative analyses (Thierry, 2021, 2007), we organize these traits into three core dimensions: socio-cognitive demands, behavioral inhibition, and the predictability of the social environment (Table 1). This conceptualization does not aim to redefine social tolerance itself, but rather to provide a structured basis for testing neuroanatomical hypotheses related to social style variability. It echoes recent efforts to bridge behavioral ecology and cognitive neuroscience by linking specific mental abilities – such as executive functions or metacognition – with distinct prefrontal regions shaped by social and ecological pressures (Bouret et al., 2024).

“Cross-fostering experiments (De Waal and Johanowicz, 1993), along with our own results, suggest that social tolerance grades reflect both early, possibly innate predispositions and later environmental shaping”.

(2) It would be valuable to clarify how the older individuals, especially those 20+ years old, may have influenced the observed age-related correlations (e.g., positive in grades 1-

2, negative in grades 3-4). Since primates show well-documented signs of aging, some discussion of the potential contribution of advanced age to the results could strengthen the interpretation.

We thank the reviewer for highlighting this important point. In our dataset, younger and older subjects are underrepresented, but they are distributed across all subgroups. Therefore, we do not think that it could drive the interaction effect we are reporting. In our sample, amygdala volume tended to increase with age in intolerant species and decrease in tolerant species. We included a new analysis (Figure 4) that allows providing a clearer assessment of when social grades 1 vs 4 differed in terms of amygdala and hippocampus volume. While our model accounts for age continuously, we agree that age-related variation deserves cautious interpretation and require longitudinal designs in future studies.

We also added the following statements in the discussion (lines 386-391)

“Due to a limited sample size of our study, this crossing trend, already accounted for by our continuous age model, should be further investigated. These results call for cautious interpretation of age-related variation and further emphasize the importance of longitudinal studies integrating both behavioral, cognitive and anatomical data in non-human primates, which would help to better understand the link between social environment and brain development (Song et al., 2021)”.

(3) The authors categorize the behavioral traits previously described in Thierry (2021) into 3 selfdefined cognitive requirements, however, they do not discuss under what conditions specific traits were assigned to categories or justify why these cognitive requirements were chosen. It is not fully clear from Thierry (2021) alone how each trait would align with the authors' categories. Given that these traits/categories are drawn on for their neuroanatomical hypotheses, it is important that the authors clarify this. It would be helpful to include a table with all behavioral traits with their respective categories, and explain their reasoning for selecting each cognitive requirement category.

Thank you for this important suggestion. We have extensively revised the introduction to explain how we derived from the scientific literature the three cognitive dimensions—socio-cognitive demands, behavioral inhibition, and predictability of the social environment—. We now provide a complete overview of the 18 behavioral traits described in Thierry's framework and their cognitive classification in a dedicated table, along with hypothesized neural correlates. We have also mentioned traits that were not classified in our framework along with short justification of this classification. We believe this addition significantly improves the transparency and intelligibility of our conceptual approach.

“The concept of social tolerance, central to this comparative approach, has sometimes been used in a vague or unidimensional way. As Bernard Thierry (2021) pointed out, the notion was initially constructed around variations in agonistic relationships – dominance, aggressiveness, appeasement or reconciliation behaviors – before being expanded to include affiliative behaviors, allomaternal care or male–male interactions (Thierry, 2021). These traits do not necessarily align along a single hierarchical axis but rather reflect a multidimensional complexity of social style, in which each trait may have co-evolved with others (Thierry, 2021, 2000; Thierry et al., 2004). Moreover, the lack of a standardized scientific definition has sometimes led to labeling species as “tolerant” or “intolerant” without explicit criteria (Gumert and Ho, 2008; Patzelt et al., 2014). These behavioral differences are characterized by different styles of dominance (Balasubramaniam et al., 2012), severity of agonistic interactions (Duboscq et al., 2014), nepotism (Berman and Thierry, 2010; Duboscq et al., 2013; Sueur et al., 2011) and submission signals (De Waal and Luttrell, 1985; Rincon et al., 2023), among the 18 covariant behavioral traits described in Thierry's classification of social tolerance (Thierry, 2021, 2017, 2000)”.

“To ground the investigation of social tolerance in a comparative neuroanatomical framework, we introduce a tentative working model that articulates behavioral traits, cognitive dimensions, and their potential subcortical neural substrates. Drawing upon 18 behavioral traits identified in Thierry’s comparative analyses (Thierry, 2021, 2007), we organized these traits into three core dimensions: socio-cognitive demands, behavioral inhibition, and the predictability of the social environment (Table 1). This conceptualization does not aim to redefine social tolerance itself, but rather to provide a structured basis for testing neuroanatomical hypotheses related to social style variability. It echoes recent efforts to bridge behavioral ecology and cognitive neuroscience by linking specific mental abilities – such as executive functions or metacognition – with distinct prefrontal regions shaped by social and ecological pressures (Bouret et al., 2024; Testard 2022)”.

(4) One of the main distinctions the authors make between high social tolerance species and low tolerance species is the level of complex socio-cognitive demands, with more tolerant species experiencing the highest demands. However, socio-cognitive demands can also be very complex for less tolerant species because they need to strategically balance behaviors in the presence of others. The relationships between socio-cognitive demands and social tolerance grades should be viewed in a more nuanced and context-specific manner.

We fully agree and we did not mean that intolerant species lives in a ‘simple’ social environment but that the ones of more tolerant species is markedly more demanding. Evidence supporting this statement include their more efficient social networks (Sueur et al., 2011) and more complex communicative skills (e.g. tolerant macaques displayed higher levels of vocal diversity and flexibility than intolerant macaques in social situation with high uncertainty (Rebout et al., 2020).

In the revised version (lines 106-122), we now highlight that socio-cognitive challenges arise across the tolerance spectrum, including in less tolerant species where strategic navigation of rigid hierarchies and risk-prone interactions is required. We hope that this addition offers a more balanced and nuanced framing of socio-cognitive demands across macaque societies

“The first category, socio-cognitive demands, refers to the cognitive resources needed to process, monitor, and flexibly adapt to complex social environments. Linking those parameters to neurological data is at the core of the social brain theory to explain the expansion of the neocortex in primates (Dunbar). Macaques social systems require advanced abilities in social memory, perspective-taking, and partner evaluation (Freeberg et al., 2012). This is particularly true in tolerant species, where the increased frequency and diversity of interactions may amplify the demands on cognitive tracking and flexibility. Tolerant macaque species typically live in larger groups with high interaction frequencies, low nepotism, and a wider range of affiliative and cooperative behaviors, including reconciliation, coalition-building, and signal flexibility (REF). Tolerant macaque species also exhibit a more diverse and flexible vocal and facial repertoire than intolerant ones which may help reduce ambiguity and facilitate coordination in dense social networks (Rincon et al., 2023; Scopa and Palagi, 2016; Rebout 2020). Experimental studies further show that macaques can use facial expressions to anticipate the likely outcomes of social interactions, suggesting a predictive function of facial signals in managing uncertainty (Micheletta et al., 2012; Waller et al., 2016). Even within less tolerant species, like *M. mulatta*, individual variation in facial expressivity has been linked to increased centrality in social networks and greater group cohesion, pointing to the adaptive value of expressive signaling across social styles (Whitehouse et al., 2024)”.

(5) While the limitations section touches on species-related considerations, the issue of individual variability within species remains important. Given that amygdala volume can be influenced by factors such as social rank and broader life experience, it might be

useful to further emphasize that these factors could introduce meaningful variation across individuals. This doesn't detract from the current findings but highlights the importance of considering life history and context when interpreting subcortical volumes-particularly in future studies.

We have now emphasized this point in the limitations section (lines 441-456). While our current dataset does not allow us to fully control for individual-level variables across all collection centers, we recognize that factors such as rank, social exposure, and individual life history may influence subcortical volumes

“Although we explained some interspecies variability, adding subjects to our database will increase statistical power and will help addressing potential confounding factors such as age or sex in future studies. One will benefit from additional information about each subject. While considered in our modelling, the social living and husbandry conditions of the individuals in our dataset remain poorly documented. The living environment has been considered, and the size of social groups for certain individuals, particularly for individuals from the CdP, have been recorded. However, these social characteristics have not been determined for all individuals in the dataset. As previously stated, the social environment has a significant impact on the volumetry of certain regions. Furthermore, there is a lack of data regarding the hierarchy of the subjects under study and the stress they experience in accordance with their hierarchical rank and predictability of social outcomes position (McCowan et al., 2022)”.

Reviewer #2 (Public review):

We thank Reviewer #2 for its thoughtful remarks and for acknowledging the value of our comparative approach despite its inherent constraints.

Summary:

This comparative study of macaque species and the type of social interaction is both ambitious and inevitably comes with a lot of caveats. The overall conclusion is that more intolerant species have a larger amygdala. There are also opposing development profiles regarding amygdala volume depending on whether it is a tolerant or intolerant species.

To achieve any sort of power, they have combined data from 4 centres, which have all used different scanning methods, and there are some resolution differences. The authors have also had to group species into 4 classifications - again to assist with any generalisations and power. They have focused on the volumes of two structures, the amygdala and the hippocampus, which seems appropriate. Neither structure is homogeneous and so it may well be that a targeted focus on specific nuclei or subfields would help (the authors may well do this next) - but as the variables would only increase further along with the number of potential comparisons, alongside small group numbers, it seems only prudent to treat these findings as preliminary. That said, it is highly unlikely that large numbers of macaque brains will become available in the near future.

This introduction is by way of saying that the study achieves what it sets out to do, but there are many reasons to see this study as preliminary. The main message seems to be twofold: (1) that more intolerant species have relatively larger amygdalae, and (2) that with development, there is an opposite pattern of volume change (increasing with age in intolerant species and decreasing with age in tolerant species). Finding 1 is the opposite of that predicted in Table 1 - this is fine, but it should be made clearer in the Discussion that this is the case, otherwise the reader may feel confused. As I read it, the authors have switched their prediction in the Discussion, which feels uncomfortable.

We thank the reviewer for this important observation. In the original version, Table 1 presented simplified direct predictions linking social tolerance grades to amygdala and hippocampus volumes. We recognize that this formulation may have created confusion. In the revised manuscript, we have thoroughly restructured the table and its accompanying rationale. Table 1 now better reflects our conceptual framework grounded in three cognitive dimensions—sociocognitive demands, behavioral inhibition, and social predictability—each linked to behavioral traits and associated neural hypotheses based on published literature. This updated framework, detailed in lines 144-169 of the introduction, provides a more nuanced basis for interpreting our results and avoids the inconsistencies previously noted. The Discussion was also revised accordingly (lines 329-255) to clarify where our findings diverge from the original predictions and to explore alternative explanations based on social complexity. Rather than directly predicting amygdala size from social tolerance grades, we propose that variation in volume emerges from differing combinations of cognitive pressures across species.

It is inevitable that the data in a study of this complexity are all too prone to post hoc considerations, to which the authors indulge. In the case of Grade 1 species, the individuals have a lot to learn, especially if they are not top of the hierarchy, but at the same time, there are fewer individuals in the troop, making predictions very tricky. As noted above, I am concerned by the seemingly opposite predictions in Table 1 and those in the Discussion regarding tolerance and amygdala volume. (It may be that the predictions in Table 1 are the opposite of how I read them, in which case the Table and preceding text need to align.)

In order to facilitate the interpretation of our Bayesian modelling, we have selected a more focused ROI in our automatic segmentation procedure of the Hippocampus (from Hippocampal Formation to Hippocampus) and have added to the new analysis (Figure 4) that helps to properly test whether the hippocampus significantly differs between species from social grade 1 vs 4. The present analysis found that this is the case in adult monkeys. This is therefore consistent with our hypothesis that amygdala volumes are principally explained by heightened sociocognitive demands in more tolerant species.

We also acknowledge the reviewer's concerns about the limited generalizability due to our sample. The challenges of comparative neuroimaging in non-human primates—especially when using post-mortem datasets—are substantial. Given the ethical constraints and the rarity of available specimens, increasing the number of individuals or species is not feasible in the short term. However, we have made all data and code publicly available and clearly stated the limitations of our sample in the manuscript. Despite these constraints, we believe our dataset offers an unprecedented comparative perspective, particularly due to the inclusion of rare and tolerant species such as *M. tonkeana*, *M. nigra*, and *M. thibetana*, which have never been included in structural MRI studies before. We hope this effort will serve as a foundation for future collaborative initiatives in primate comparative neuroscience.

Reviewer #3 (Public review):

We thank Reviewer #3 for their thoughtful and detailed review. Their comments helped us refine both the conceptual and interpretative aspects of the manuscript. We respond point by point below.

Summary:

*In this study, the authors were looking at neurocorrelates of behavioural differences within the genus *Macaca*. To do so, they engaged in real-world dissection of dead animals (unconnected to the present study) coming from a range of different institutions. They subsequently compare different brain areas, here the amygdala and the hippocampus, across species. Crucially, these species have been sorted according to*

*different levels of social tolerance grades (from 1 to 4). 12 species are represented across 42 individuals. The sampling process has weaknesses ("only half" of the species contained by the genus, and *Macaca mulatta*, the rhesus macaque, representing 13 of the total number of individuals), but also strengths (the species are decently well represented across the 4 grades) for the given purpose and for the amount of work required here. I will not judge the dissection process as I am not a neuroanatomist, and I will assume that the different interventions do not alter volume in any significant ways / or that the different conditions in which the bodies were kept led to the documented differences across species.*

25 brains were extracted by the authors themselves who are highly with this procedure. Overall, we believe that dissection protocols did not alter the total brain volume. Despite our expertise, we experienced some difficulties to not damage the cerebellum. Therefore, this region was not included in our analysis. We also noted that this brain region was also damaged or absent from the Prime-DE dataset.

Several protocols were used to prepare and store tissue. It could have impacted the total brain volume.

We agree that differences in tissue preparation and storage could potentially affect total brain volume. Therefore, we explicitly included the main sample preparation variable — whether brains had been previously frozen — as a covariate in our model. This factor did not explain our results. Moreover, Figures 1D and 1I display the frozen status and its correlation with the amygdala and hippocampus ratios, respectively. Figure 2 shows the parameters of the model and the posterior distributions for the frozen status and total brain volume effects.

There are two main results of the study. First, in line with their predictions, the authors find that more tolerant macaque species have larger amygdala, compared to the hippocampus, which remains undifferentiated across species. Second, they also identify developmental effects, although with different trends: in tolerant species, the amygdala relative volume decreases across the lifespan, while in intolerant species, the contrary occurs. The results look quite strong, although the authors could bring up some more clarity in their replies regarding the data they are working with. From one figure to the other, we switch from model-calculated ratio to modelpredicted volume. Note that if one was to sample a brain at age 20 in all the grades according to the model-predicted volumes, it would not seem that the difference for amygdala would differ much across grades, mostly driven with Grade 1 being smaller (in line with the main result), but then with Grade 2 bigger than Grade 3, and then Grade 4 bigger once again, but not that different from Grade 2.

Overall, despite this, I think the results are pretty strong, the correlations are not to be contested, but I also wonder about their real meaning and implications. This can be seen under 3 possible aspects:

(1) Classification of the social grade

While it may be familiar to readers of Thierry and collaborators, or to researchers of the macaque world, there is no list included of the 18 behavioral traits used to define the three main cognitive requirements (socio-cognitive demands, predictability of the environment, inhibitory control). It would be important to know which of the different traits correspond to what, whether they overlap, and crucially, how they are realized in the 12 study species, as there could be drastic differences from one species to the next. For now, we can only see from Table S1 where the species align to, but it would be a good addition to have them individually matched to, if not the 18 behavioral traits, at least the 3 different broad categories of cognitive requirements.

We fully agree with this observation. In the revised version of the manuscript, we now include a detailed conceptual table listing all 18 behavioral traits from Thierry's framework. For each trait, we provide its underlying social implications, its associated cognitive dimension (when applicable), and the hypothesized neural correlate.

While some traits may could have been arguably classified in several cognitive dimensions (e.g. reconciliation rate), we preferred to assign each to a unique dimension for clarity. Additionally, the introduction (lines 95-169 + Table1) now explains how each trait was evaluated based on existing literature and assigned to one of the three proposed cognitive categories: socio-cognitive demands, behavioral inhibition, or social unpredictability. This structure offers a clearer and more transparent basis for the neuroanatomical hypotheses tested in the study.

“Navigating social life in primate societies requires substantial cognitive resources: individuals must not only track multiple relationships, but also regulate their own behavior, anticipate others' reactions, and adapt flexibly to changing social contexts. Taken advantage of databases of magnetic resonance imaging (MRI) structural scans, we conducted the first comparative study integrating neuroanatomical data and social behavioral data from closely related primate species of the same genus to address the following questions: To what extent can differences in volumes of subcortical brain structures be correlated with varying degrees of social tolerance? Additionally, we explored whether these dispositions reflect primarily innate features, shaped by evolutionary processes, or acquired through socialization within more or less tolerant social environments”.

“The first category, socio-cognitive demands, refers to the cognitive resources needed to process, monitor, and flexibly adapt to complex social environments. Linking those parameters to neurological data is at the core of the social brain theory to explain the expansion of the neocortex in primates (Dunbar). Macaques social systems require advanced abilities in social memory, perspective-taking, and partner evaluation (Freeberg et al., 2012). This is particularly true in tolerant species, where the increased frequency and diversity of interactions may amplify the demands on cognitive tracking and flexibility. Tolerant macaque species typically live in larger groups with high interaction frequencies, low nepotism, and a wider range of affiliative and cooperative behaviors, including reconciliation, coalition-building, and signal flexibility (REF). Tolerant macaque species also exhibit a more diverse and flexible vocal and facial repertoire than intolerant ones which may help reduce ambiguity and facilitate coordination in dense social networks (Rincon et al., 2023; Scopa and Palagi, 2016; Rebout 2020). Experimental studies further show that macaques can use facial expressions to anticipate the likely outcomes of social interactions, suggesting a predictive function of facial signals in managing uncertainty (Micheletta et al., 2012; Waller et al., 2016). Even within less tolerant species, like *M. mulatta*, individual variation in facial expressivity has been linked to increased centrality in social networks and greater group cohesion, pointing to the adaptive value of expressive signaling across social styles (Whitehouse et al., 2024)”.

“The second category, inhibitory control, includes traits that involve regulating impulsivity, aggression, or inappropriate responses during social interactions. Tolerant macaques have been shown to perform better in tasks requiring behavioral inhibition and also express lower aggression and emotional reactivity in both experimental and natural contexts (Joly et al., 2017; Loyant et al., 2023). These features point to stronger self-regulation capacities in species with egalitarian or less rigid hierarchies. More broadly, inhibition – especially in its strategic form (self-control) – has been proposed to play a key role in the cohesion of stable social groups. Comparative analyses across mammals suggest that this capacity has evolved primarily in anthropoid primates, where social bonds require individuals to suppress immediate impulses in favour of longer-term group stability (Dunbar and Shultz, 2025). This view echoes the conjecture of Passingham and Wise (2012), who proposed that the emergence

of prefrontal area BA10 in anthropoids enabled the kind of behavioural flexibility needed to navigate complex social environments (Passingham et al., 2012)”.

“The third category, social environment predictability, reflects how structured and foreseeable social interactions are within a given society. In tolerant species, social interactions are more fluid and less kin-biased, leading to greater contextual variation and role flexibility, which likely imply a sustained level of social awareness. In fact, as suggested by recent research, such social uncertainty and prolonged incentives are reflected by stress-related physiology : tolerant macaques such as *M. tonkeana* display higher basal cortisol levels, which may be indicative of a chronic mobilization of attentional and regulatory resources to navigate less predictable social environments (Sadoughi et al., 2021)”.

“Each behavioral trait was individually evaluated based on existing empirical literature regarding the types of cognitive operations it likely involves. When a primary cognitive dimension could be identified, the trait was assigned accordingly. However, some behaviors – such as maternal protection, allomaternal care, or delayed male dispersal – do not map neatly onto a single cognitive process. These traits likely emerge from complex configurations of affective and socialmotivational systems, and may be better understood through frameworks such as attachment theory (Suomi, 2008), which emphasizes the integration of social bonding, emotional regulation, and contextual plasticity. While these dimensions fall beyond the scope of the present framework, they offer promising directions for future research, particularly in relation to the hypothalamic and limbic substrates of social and reproductive behavior”.

“Rather than forcing these traits into potentially misleading categories, we chose to leave them unclassified within our current cognitive framework. This decision reflects both a commitment to conceptual clarity and the recognition that some behaviors emerge from a convergence of cognitive demands that cannot be neatly isolated. This tripartite framework, leaving aside reproductive-related traits, provides a structured lens through which to link behavioral diversity to specific cognitive processes and generate neuroanatomical predictions”.

(2) Issue of nature vs nurture

Another way to look at the debate between nature vs nurture is to look at phylogeny. For now, there is no phylogenetic tree that shows where the different grades are realized. For example, it would be illuminating to know whether more related species, independently of grades, have similar amygdala or hippocampus sizes. Then the question will go to the details, and whether the grades are realized in particular phylogenetic subdivisions. This would go in line with the general point of the authors that there could be general species differences.

As pointed out by Thierry and collaborators, the social tolerance concept is already grounded in a phylogenetic framework as social tolerance matches the phylogenetical tree of these macaque species, suggesting a biological ground of these behavioral observations. Given the modest sample size and uneven species representation, we opted not to adopt tools such as Phylogenetic Generalized Least Squares (PGLS) in our analysis. Our primary aim in this study was to explore neuroanatomical variation as a function of social traits, not to perform a phylogenetic comparative analysis per se. That said, we now explicitly acknowledge this limitation in the Discussion and indicate that future work using larger datasets and phylogenetic methods will be essential to disentangle social effects from evolutionary relatedness. We hope that making our dataset openly available will facilitate such futures analyses.

With respect to nurture, it is likely more complicated: one needs to take into account the idiosyncrasies of the life of the individual. For example, some of the cited literature in

humans or macaques suggests that the bigger the social network, the bigger the brain structure considered. Right, but this finding is at the individual level with a documented life history. Do we have any of this information for any of the individuals considered (this is likely out of the scope of this paper to look at this, especially for individuals that did not originate from CdP)?

We appreciate this insightful observation. Indeed, findings from studies in humans and nonhuman primates showing associations between brain structure and social network size typically rely on detailed life history and behavioral data at the individual level. Unfortunately, such finegrained information was not consistently available across our entire sample. While some individuals from the Centre de Primatologie (CdP) were housed in known group compositions and social settings, we did not have access to longitudinal social data—such as rank, grooming rates, or network centrality—that would allow for robust individual-level analyses. We now acknowledge this limitation more clearly in the Discussion (lines 436-443), and we fully agree that future work combining neuroimaging with systematic behavioral monitoring will be necessary to explore how species-level effects interact with individual social experience.

(3) Issue of the discussion of the amygdala's function

The entire discussion/goal of the paper, states that the amygdala is connected to social life. Yet, before being a "social center", the amygdala has been connected to the emotional life of humans and non-humans alike. The authors state L333/34 that "These findings challenge conventional expectations of the amygdala's primary involvement in emotional processes and highlight the complexity of the amygdala's role in social cognition". First, there is no dichotomy between social cognition and emotion. Emotion is part of social cognition (unless we and macaques are robots). Second, there is nowhere in the paper a demonstration that the differences highlighted here are connected to social cognition differences per se. For example, the authors have not tested, say, if grade 4 species are more afraid of snakes than grade 1 species. If so, one could predict they would also have a bigger amygdala, and they would probably also find it in the model. My point is not that the authors should try to correlate any kind of potential aspect that has been connected to the amygdala in the literature with their data (see for example the nice review by DomínguezBorràs and Vuilleumier, <https://doi.org/10.1016/B978-0-12-823493-8.00015-8>), but they should refrain from saying they have challenged a particular aspect if they have not even tested it. I would rather engage the authors to try and discuss the amygdala as a multipurpose center, that includes social cognition and emotion.

We thank the reviewer for this important and nuanced point. We have revised the manuscript to adopt a more cautious and integrative tone regarding the function of the amygdala. In the revised Discussion (lines 341-355), we now explicitly state that the amygdala is involved in a broad range of processes—emotional, social, and affective—and that these domains are deeply intertwined. Rather than proposing a strict dissociation, we now suggest that the amygdala supports integrated socio-emotional functions that are mobilized differently across social tolerance styles. We also cite recent relevant literature (e.g., Domínguez-Borràs & Vuilleumier, 2021) to support this view and have removed any claim suggesting we challenge the emotional function of the amygdala per se. Our aim is to contribute to a richer understanding of how affective and social processes co-construct structural variation in this region.

Strengths:

Methods & breadth of species tested.

Weaknesses:

Interpretation, which can be described as 'oriented' and should rather offer additional views.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Private Comments:

(1) Table 1 should be formatted for clarity i.e., bolded table headers, text realignment, and spacing. It was not clear at first glance how information was organized. It may also be helpful to place behavioral traits as the first column, seeing that these traits feed into the author's defined cognitive requirements.

We have reformatted Table 1 to improve clarity and readability. Behavioral traits now appear in the first column, followed by cognitive dimensions and hypothesized neural correlates. Column headers have been bolded and alignment has been standardized.

(2) Figures could include more detail to help with interpretations. For example, Figure 3 should define values included on the x-axis in the figure caption, and Figure 4 should explain the use of line, light color, and dark color. Figure 1 does not have a y-axis title.

The figures have been revised and legends completed to ensure more clarity.

(3) Please proofread for typos throughout.

The manuscript has been carefully proofread, and all typographical and grammatical errors have been corrected. These changes are visible in the tracked version.

Reviewer #2 (Recommendations for the authors):

Specific comments:

(1) Given all of the variability would it not be a good idea to just compare (eg in the supplemental) the macaque data from just the Strasbourg centre for m mulatta and m toneanna. I appreciate the ns will be lower, but other matters are more standardized.

We fully understand the reviewer's suggestion to restrict the comparison to data collected at a single site in order to minimize inter-site variability. However, as noted, such an analysis would come at the cost of statistical power, as the number of individuals per species within a single center is small. For example, while *M. tonkeana* is well represented at the Strasbourg centre, only one individual of *M. mulatta* is available from the same site. Thus, a restricted comparison would severely limit the interpretability of results, particularly for age-related trajectories. To address variability, we included acquisition site and brain preservation method as covariates or predictors where appropriate, and we have been cautious in our interpretations. We also now emphasize in the Methods and Discussion the value of future datasets with more standardized acquisition protocols across species and centers. We hope that by openly sharing our data and workflow, we can contribute to this broader goal.

(2) I have various minor edits:

(a) L 25 abstract - Specify what is meant by 'opposite trend'; the reader cannot infer what this is.

Modified in line 25-28: "Unexpectedly, tolerant species exhibited a decrease in relative amygdala volume across the lifespan, contrasting with the age-related increase observed in intolerant species—a developmental pattern previously undescribed in primates."

(b) L67 - The reference 'Manyprimates' needs fixing as it does in the references section.

After double checking, Manyprimates studies are international collaborative efforts that are supposed to be cite this way (<https://manyprimates.github.io/#pubs>).

(c) L74 - Taking not Taken.

This typo has been corrected.

(d) L129 - It says 'total volume', but this is corrected total volume?

We have clarified in the figures legends that the “total brain volume” used in our analyses excludes the cerebellum and the myelencephalon, as specified in our image preprocessing protocol. This ensures consistency across individuals and institutions.

(e) L138 - Suddenly mentions 'frozen condition' without any prior explanation - this needs explaining in the legend - also L144.

We have added an explanation of the ‘frozen condition’ variable in in the relevant figure legend.

(f) L166 - Results - it would be helpful to remind readers what Grade 1 signifies, ie intolerant species.

We now include a brief reminder in the Results section that Grade 1 corresponds to socially intolerant species, to help readers unfamiliar with the classification (Lines 240-251).

(g) Figure 4 - Provide the ns for each of the 4 grades to help appreciate the meaningfulness of the curves, etc.

The number of subjects has been added to the Figure and a novel analysis helps in the revised ms help to appreciate the meaningfulness of some of these curves.

(h) L235 - 'we had assumed that species of high social tolerance grade would have presented a smaller amygdala in size compared to grade 1'. But surely this is the exact opposite of what is predicted in Table 1 - ie, the authors did not predict this as I read the paper (Unless Table 1 is misleading/ambiguous and needs clarification).

As discussed in our response to Reviewer #2 and #3, we have restructured both Table 1 and the Discussion to ensure consistency. We now explicitly state that the findings diverge from our initial inhibitory-control-based prediction and propose alternative interpretations based on sociocognitive demands.

(i) L270 - 'This observation' which?? Specify.

We have replaced ‘this observation’ with a precise reference to the observed developmental decrease in amygdala volume in tolerant species.

(j) L327 - 'groundbreaking' is just hype given that there are so many caveats - I personally do not like the word - novel is good enough.

We have replaced the word ‘groundbreaking’ with ‘novel’ to adopt a more measured and appropriate tone in the discussion.

(3) I might add that I am happy with the ethics regarding this study.

Thanks, we are also happy that we were able to study macaque brains from different species using opportunistic samplings along with already available data. We are collectively making progress on this!

(4) Finally, I should commend the authors on all the additional information that they provide re gender/age/species. Given that there are 2xs as many females as males, it would be good to know if this affects the findings. I am not a primatologist, so I don't know, for example, if the females in Grade 1 monkeys are just as intolerant as the males?

We thank the reviewer for this thoughtful comment. We now explicitly mention the female-biased sex ratio in the Methods section and report in the Results (Figure 2, Figure 3) that sex was included as a covariate in our Bayesian models. While a small effect of sex was found for hippocampal volume, no effect was observed for the amygdala. Given the strong imbalance in our dataset (2:1 female-to-male ratio), we refrained from drawing any conclusion about sex-specific patterns, as these would require larger and more balanced samples. Although we did not test for sex-by-grade interactions, we agree that this question—especially regarding whether females and males express social style differences similarly across grades—represents an important direction for future comparative work.

Reviewer #3 (Recommendations for the authors):

I found the article well-written, and very easy to follow, so I have little ways to propose improvements to the article to the authors, besides addressing the various major points when it comes to interpretation of the data.

One list I found myself wanting was in fact the list of the social tolerance grades, and the process by which they got selected into 3 main bags of socio-cognitive skills. Then it would become interesting to see how each of the 12 species compares within both the 18 grades (maybe once again out of the scope of this paper, there are likely reviews out there that already do that, but then the authors should explicitly mention so in the paper: X, 19XX have compared 15 out of 18 traits in YY number of macaque species); and within the 3 major subcognitive requirements delineated by the authors, maybe as an annex?

We thank the reviewer for this thoughtful suggestion. In the revised manuscript, we now include a detailed table (Table 1) that lists the 18 behavioral traits derived from Thierry's framework, along with their associated cognitive dimension and hypothesized neuroanatomical correlate. While we did not create a matrix mapping each of the 12 species across all 18 traits due to space and data availability constraints, we agree this is an important direction that should be tackled by primatologist. We now include a sentence (line 87-90) in the manuscript to guide readers to previous comparative reviews (e.g., Thierry, 2000; Thierry et al., 2004, 2021) that document the expression of these traits across macaque species. We also clarify that our three cognitive categories are conceptual tools intended to structure neuroanatomical predictions, and not formal clusters derived from quantitative analyses.

In the annex, it would also be good to have a general summarizing excel/R file for the raw data, with important information like age, sex, and the relevant calculated volumes for each individual. The folders available following the links do not make it an easy task for a reader to find the raw data in one place.

We fully agree with the reviewer on the importance of data accessibility. We have now uploaded an additional supplementary file in .csv format on our OSF repository, which includes individual-level metadata for all 42 macaques: species, sex, age, social grade, total brain volume, amygdala volume, and hippocampus volume. The link to this file is now explicitly mentioned in the Data Availability section. We hope this will facilitate comparisons with other datasets and improve usability for the community. In addition, we provide in a supplementary table the raw data that were used for our Bayesian modelling (see below).

The availability of the raw data would also clear up one issue, which I believe results from the modelling process: it looks odd on Figure 2, that volume ratios, defined as the

given brain area volume divided by the total brain volume, give values above 1 (especially for the hippocampus). As such, the authors should either modify the legend or the figure. In general, it would be nicer to have the "real values" somewhere easily accessible, so that they can be compared more broadly with: 1) other macaques species to address questions relevant to the species; 2) other primates to address other questions that are surely going to arise from this very interesting work!

We thank the reviewer for pointing this out. The ratio values in Figure 1 correspond to the proportion of the regional volume (amygdala or hippocampus) relative to the total brain volume, excluding the cerebellum and myelencephalon. As such, values above 0.01 (i.e., above 1% of the brain volume) are expected for these structures and do not indicate an error. We have updated the figure legend to clarify this point explicitly. In addition, we have now made a cleaned .csv file available via OSF, containing all raw volumetric data and metadata in a format that facilitates cross-species or cross-study comparisons. This replaces the previous folder-based structure, which may have been less accessible.

Typos:

L233: delete 'in'

L430: insert space in 'NMT template(Jung et al., 2021).'

<https://doi.org/10.7554/eLife.106424.2.sa0>